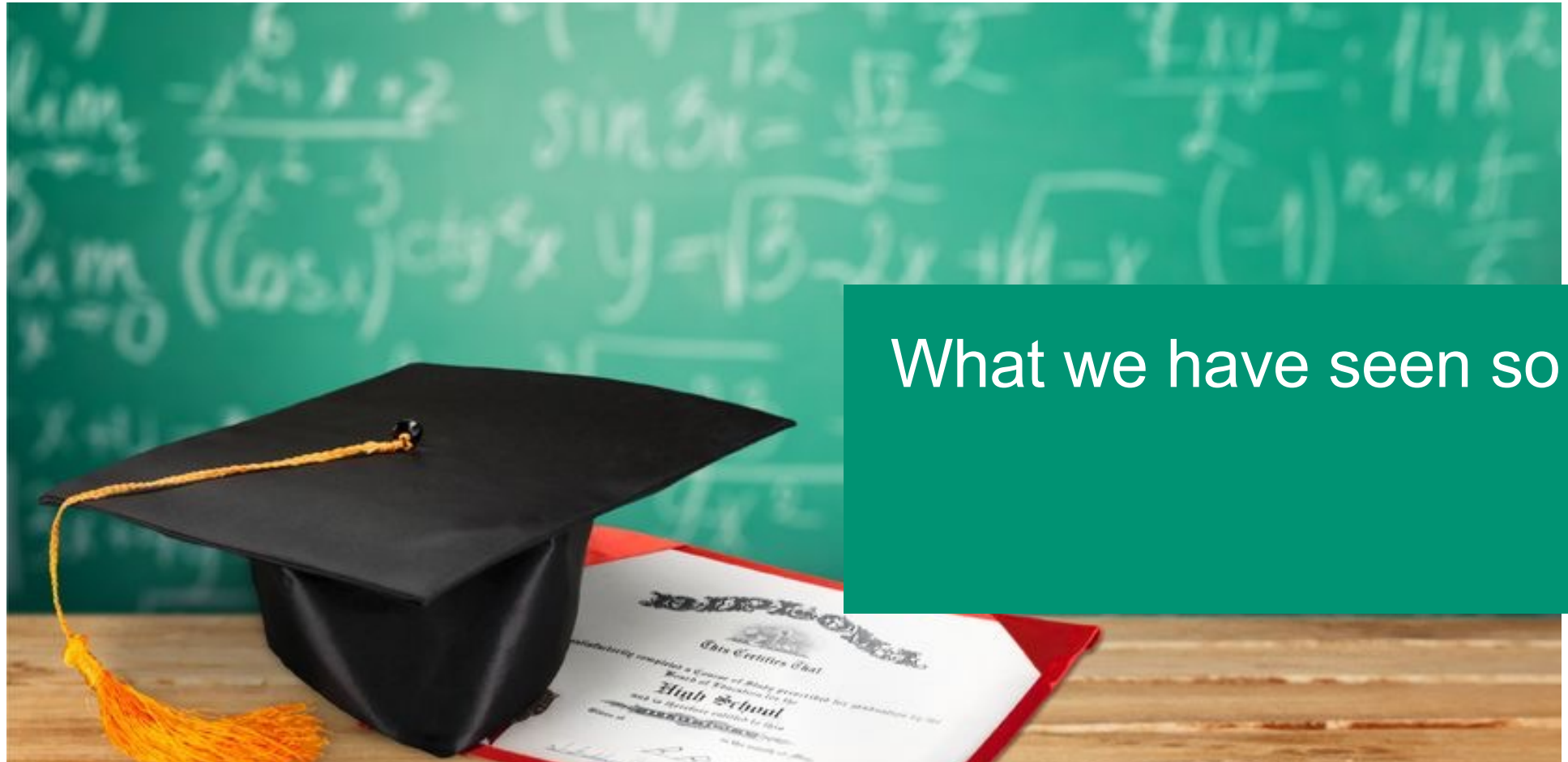


Information Retrieval

10. Probabilistic retrieval

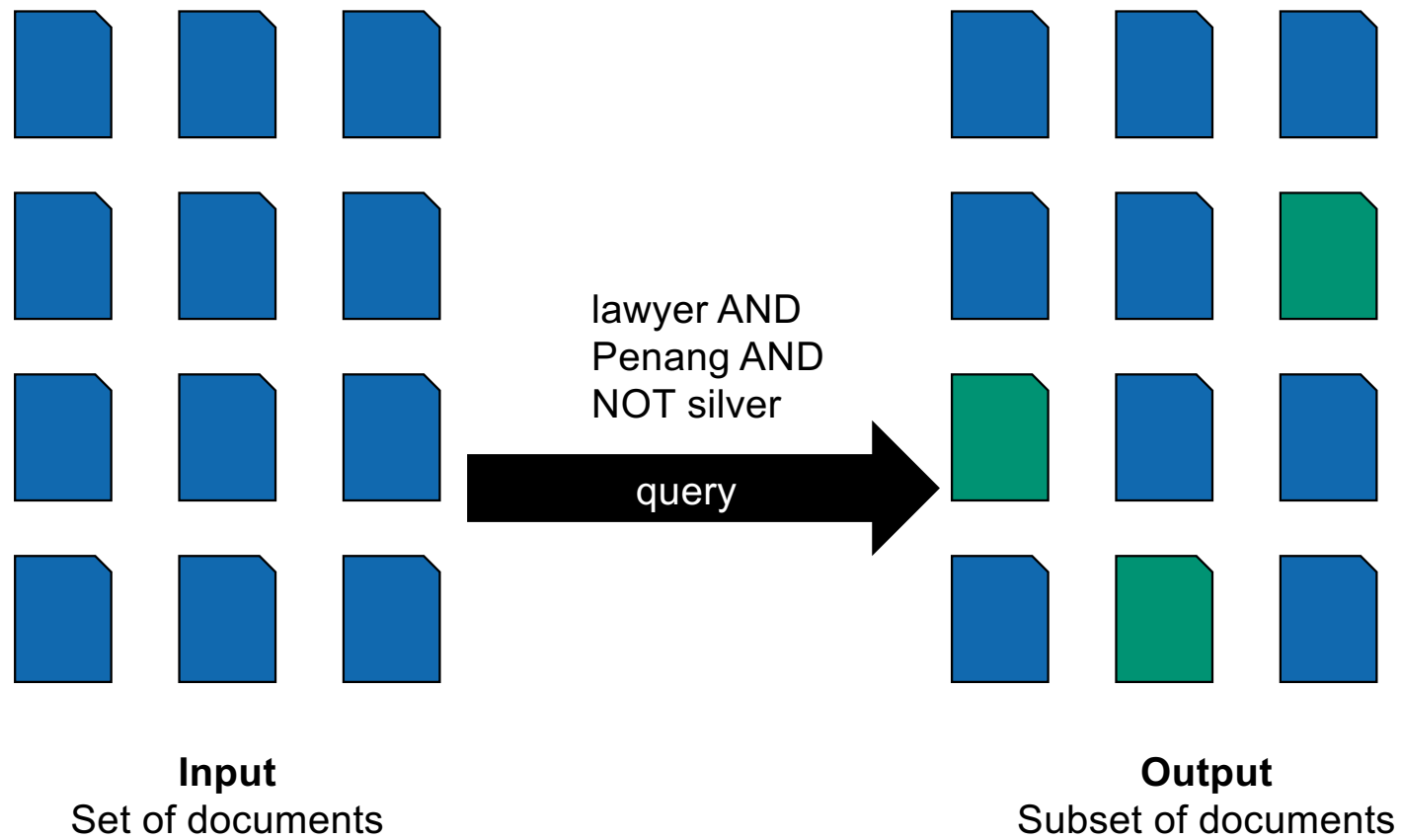
Ghislain Fourny
Spring 2022

10

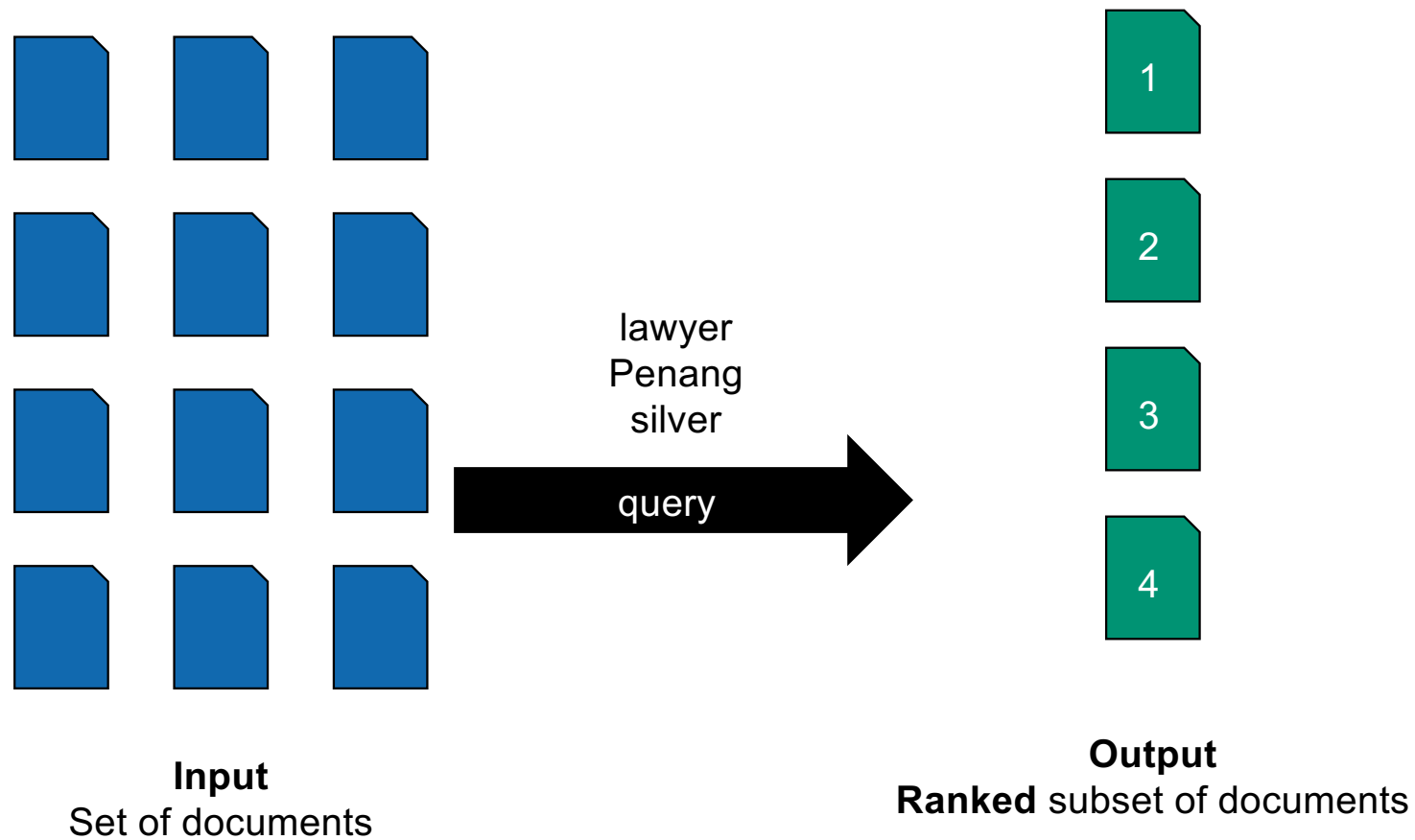


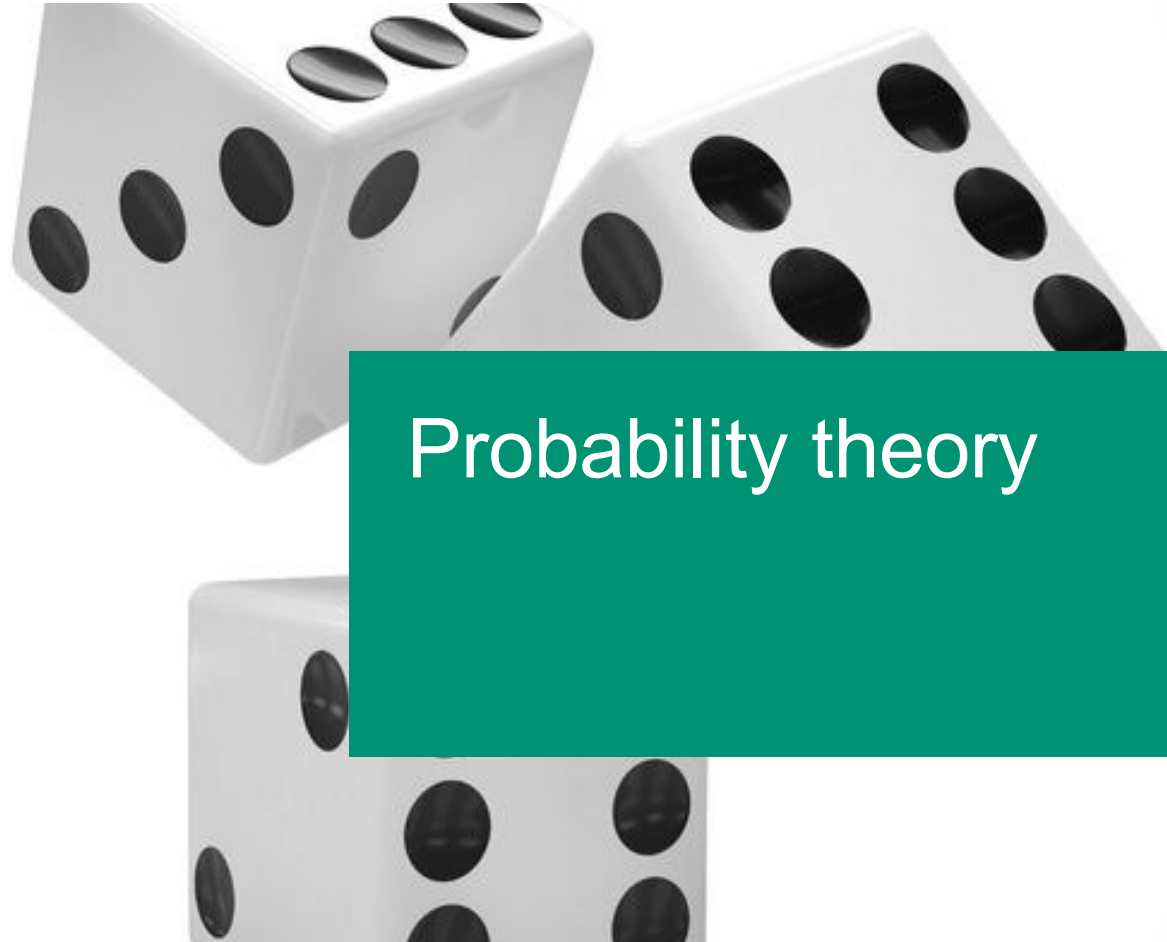
What we have seen so far

Boolean retrieval



Ranked retrieval

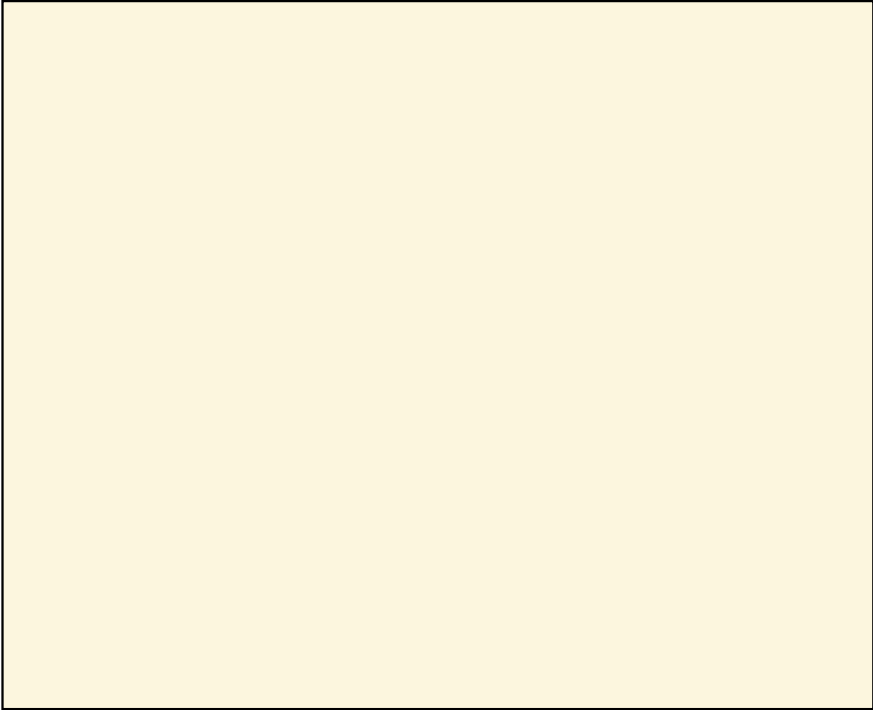




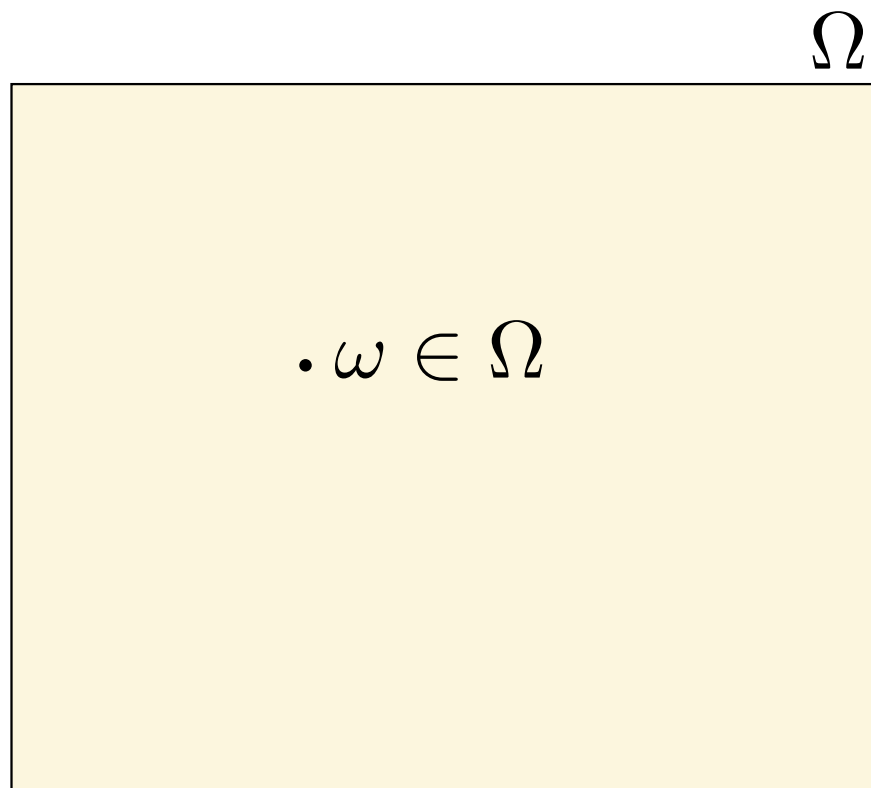
Probability theory

Universe

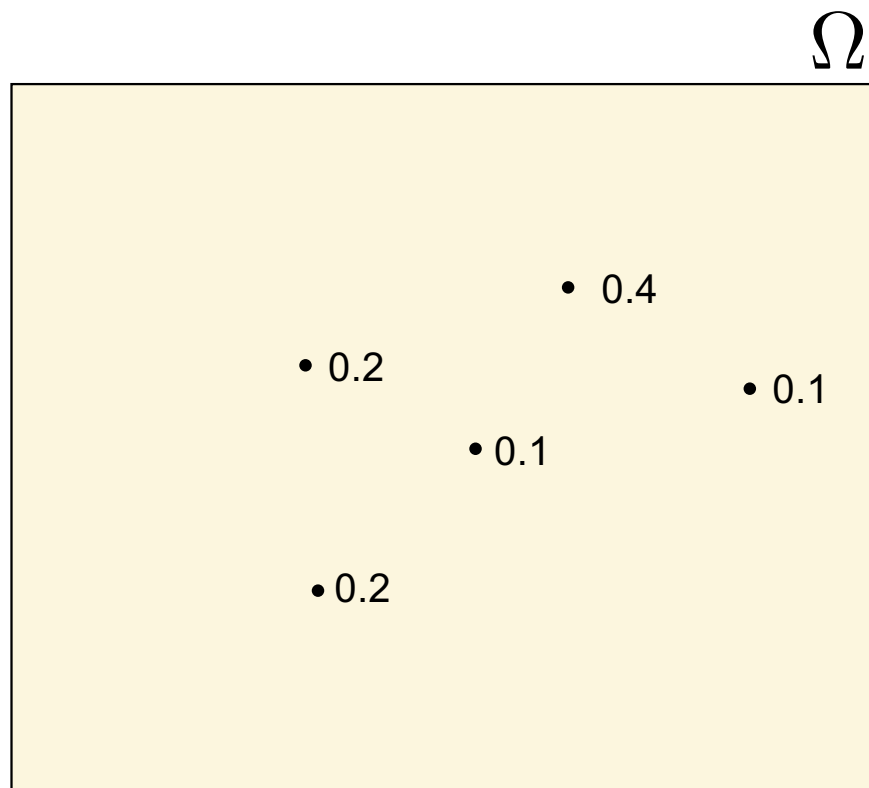
Ω



Elementary event (outcome)



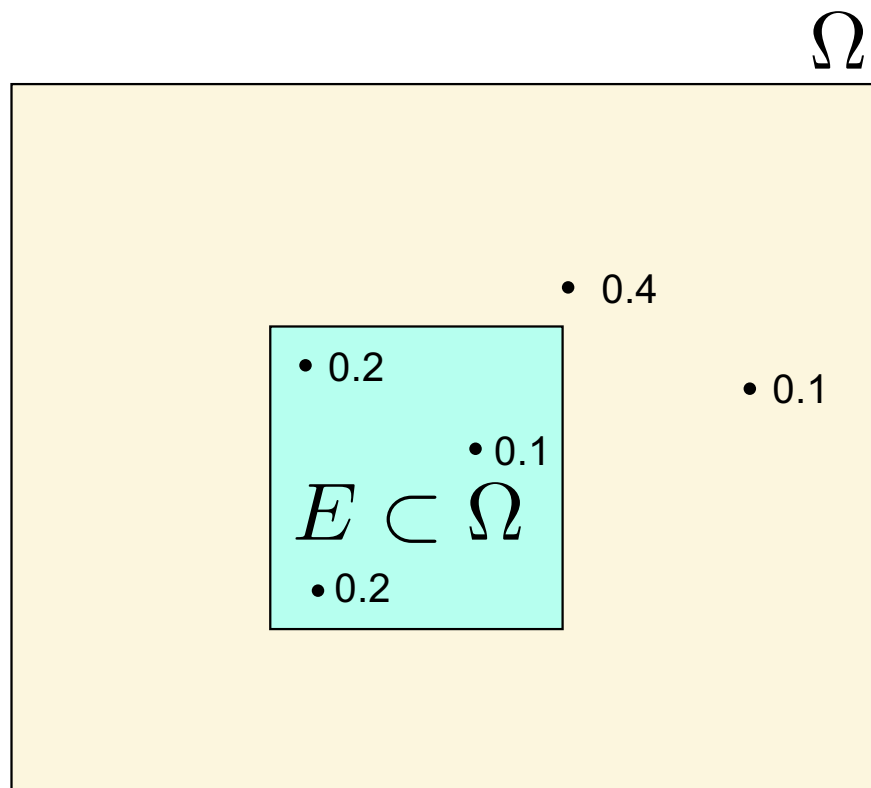
Probability distribution



$$p : \begin{array}{l} \Omega \rightarrow [0, 1] \\ \omega \mapsto p(\omega) \end{array}$$

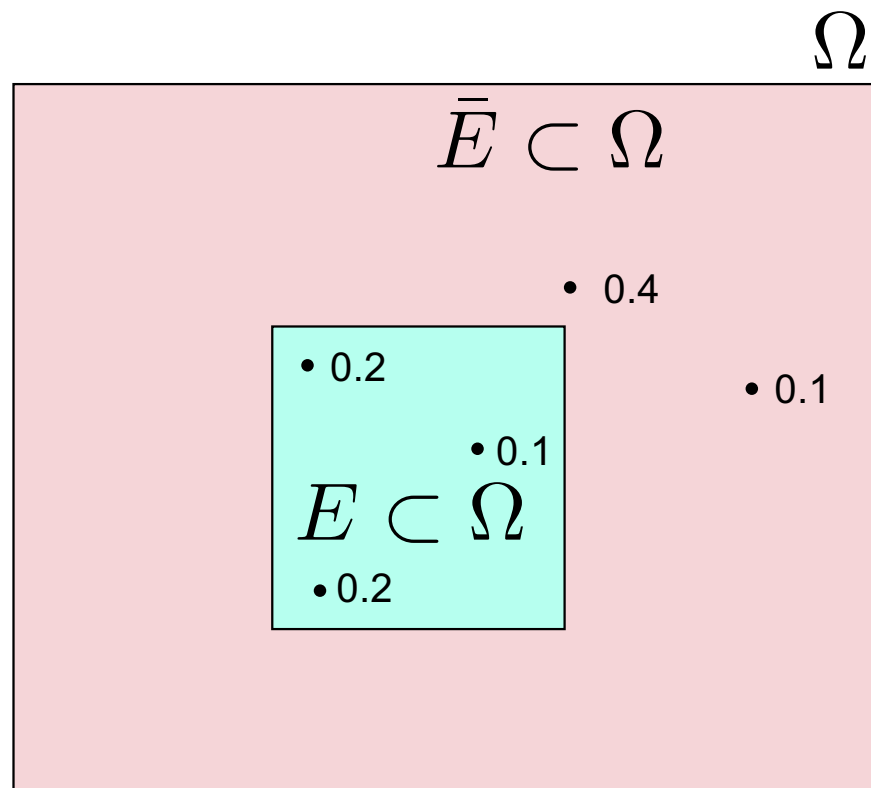
$$\sum_{\omega \in \Omega} p(\omega) = 1$$

Event



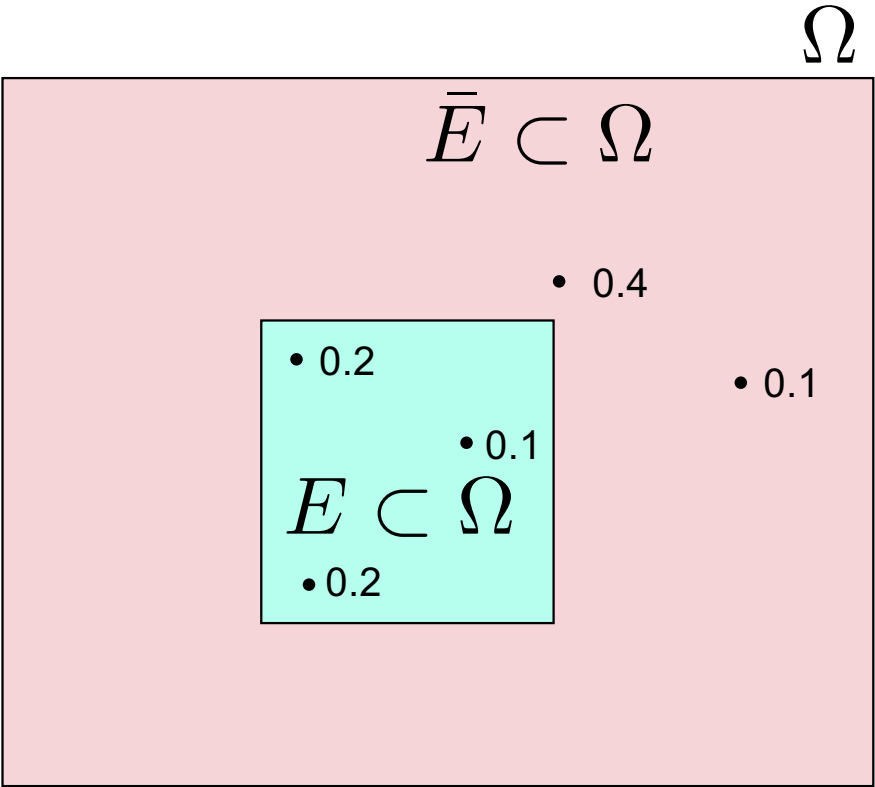
$$p(E) = \sum_{\omega \in E} p(\omega) \in [0, 1]$$

Complement



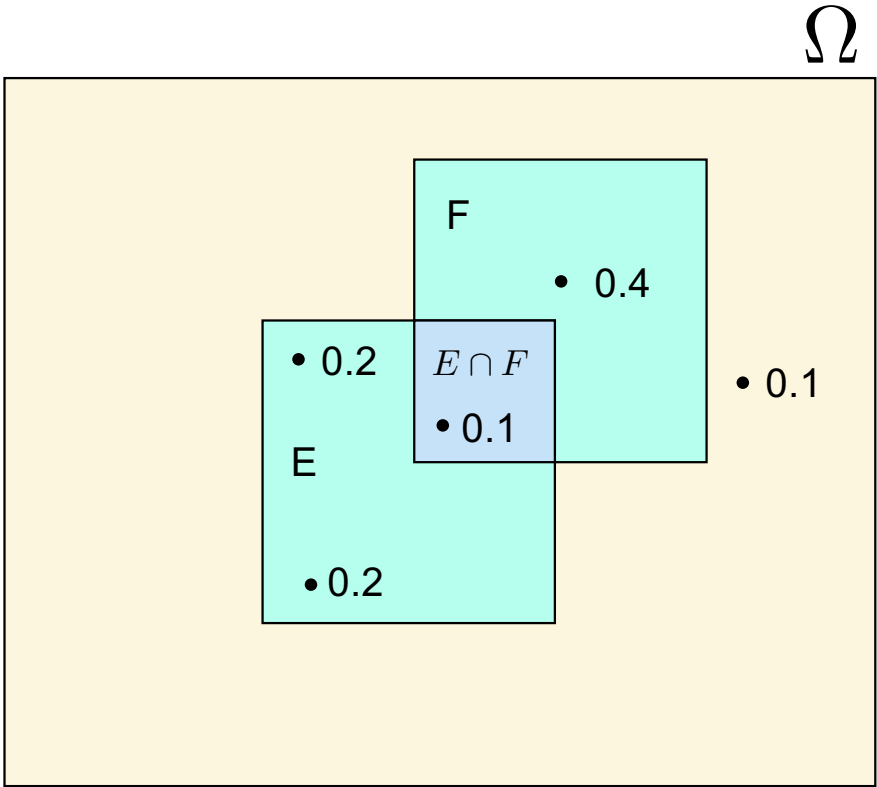
$$p(\bar{E}) = 1 - p(E)$$

Odds

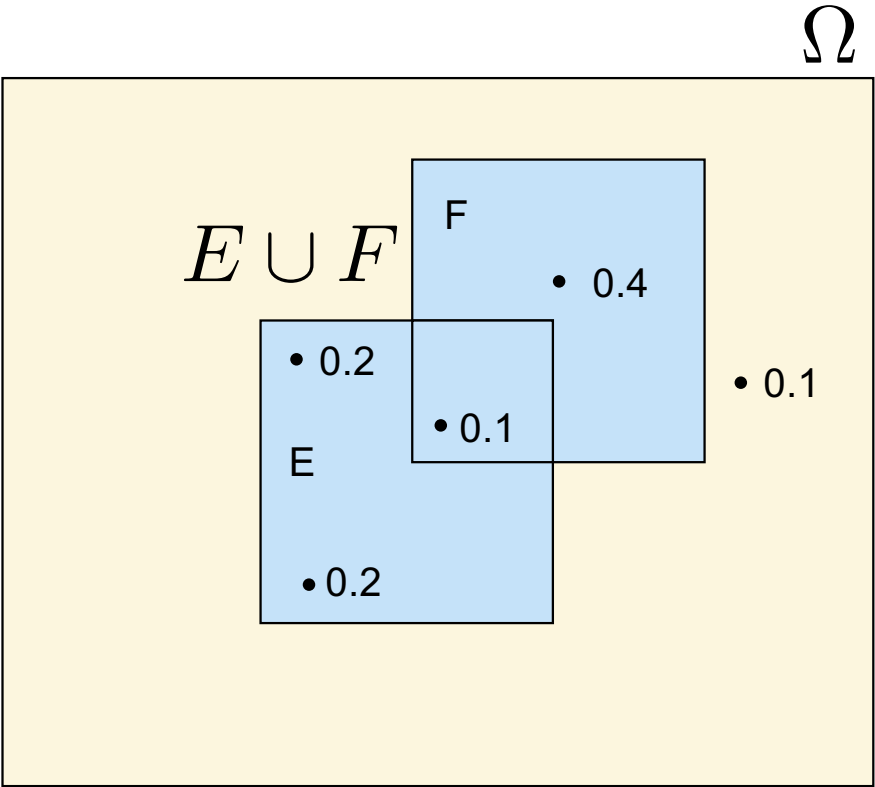


$$O_p(E) = \frac{p(E)}{p(\bar{E})}$$

Intersection

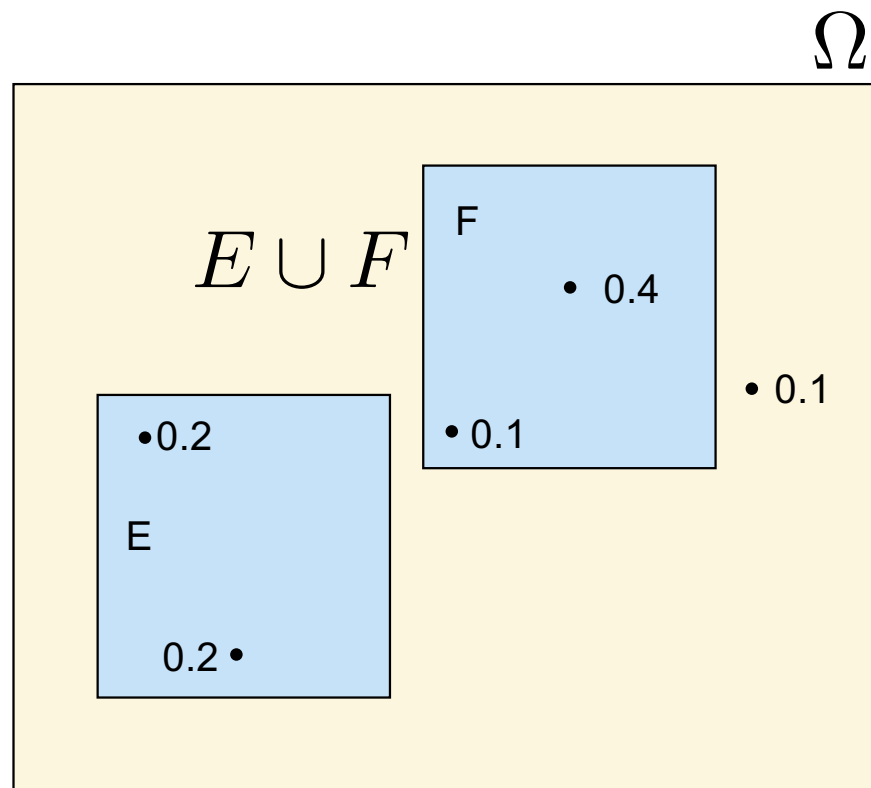


Union



$$P(E \cup F) = P(E) + P(F) - P(E \cap F)$$

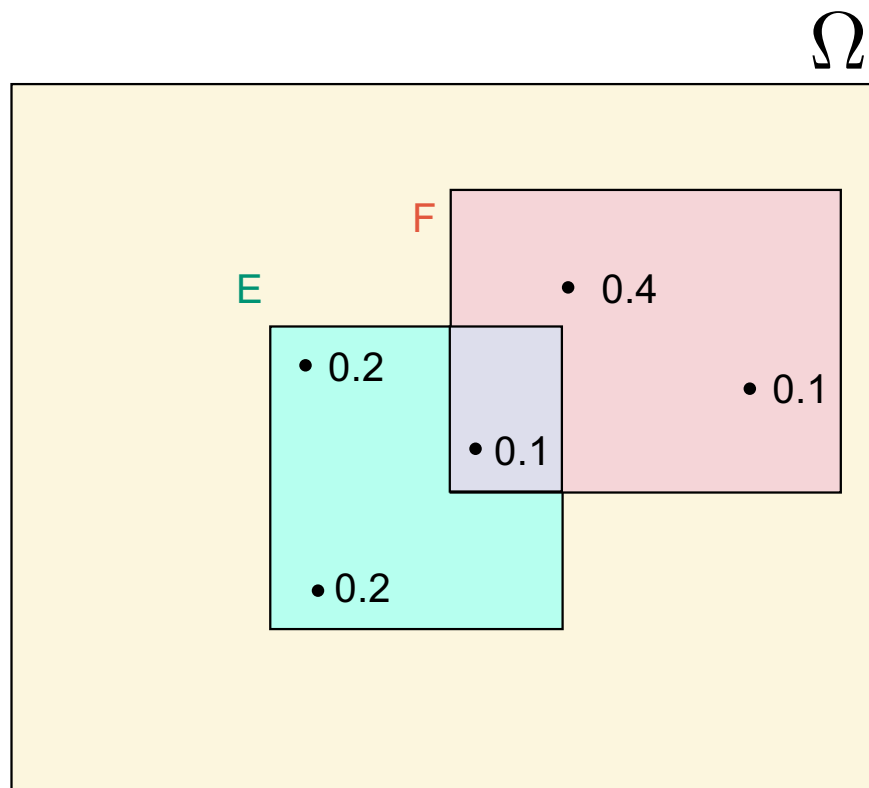
Disjoint events



$$P(E \cap F) = 0$$

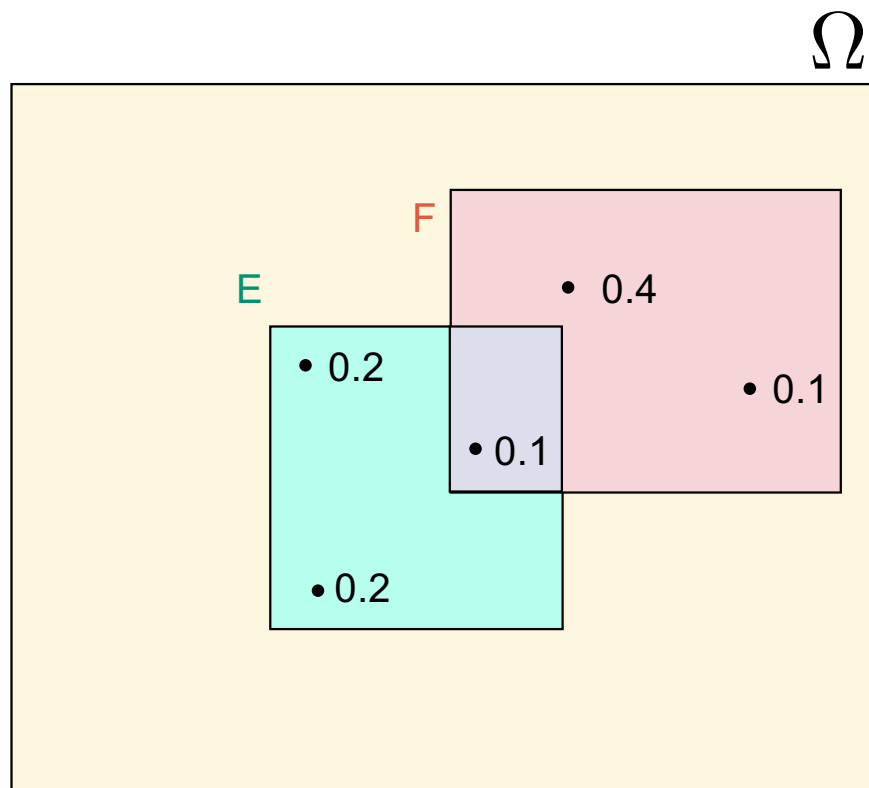
$$P(E \cup F) = P(E) + P(F)$$

Partition rule



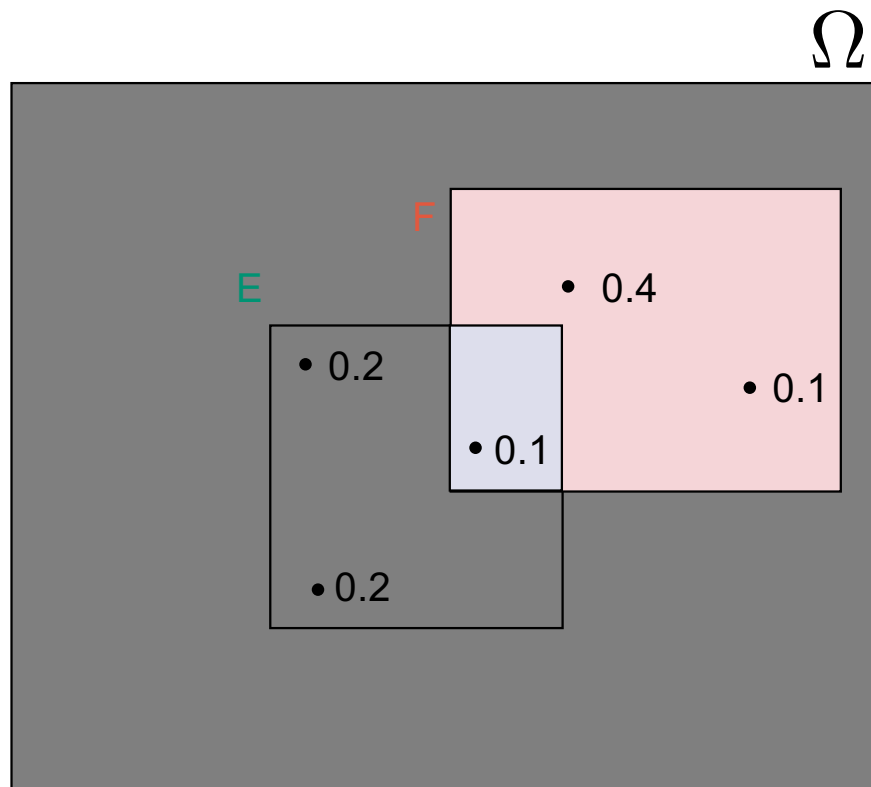
$$p(E) = p(E \cap F) + p(E \cap \bar{F})$$

Conditional probability



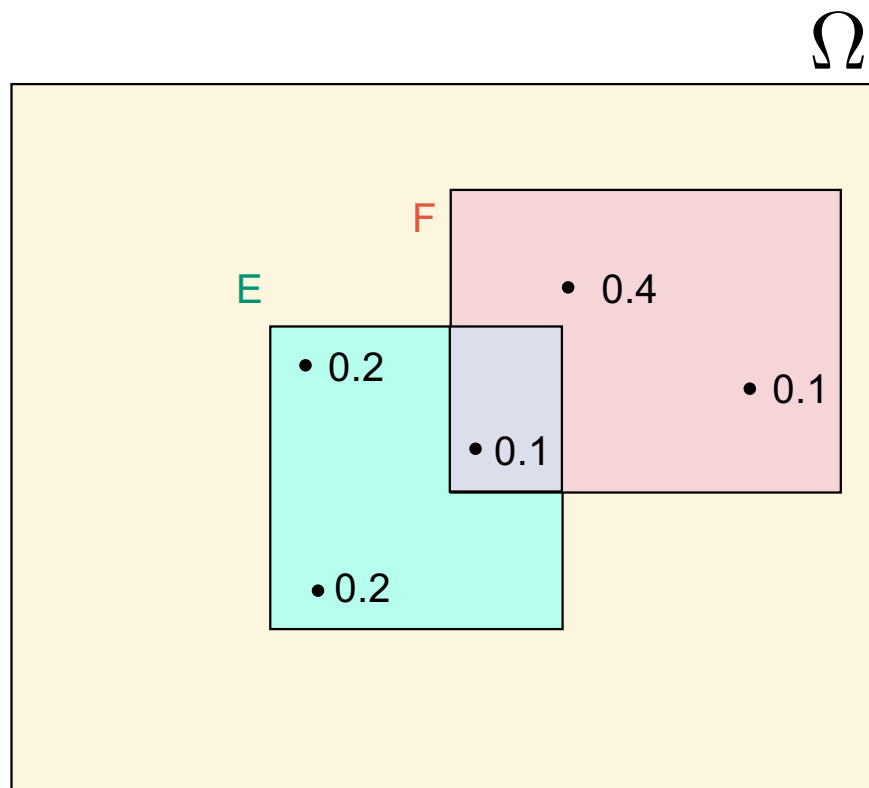
$$p(E|F) = \frac{p(E \cap F)}{p(F)}$$

Conditional probability



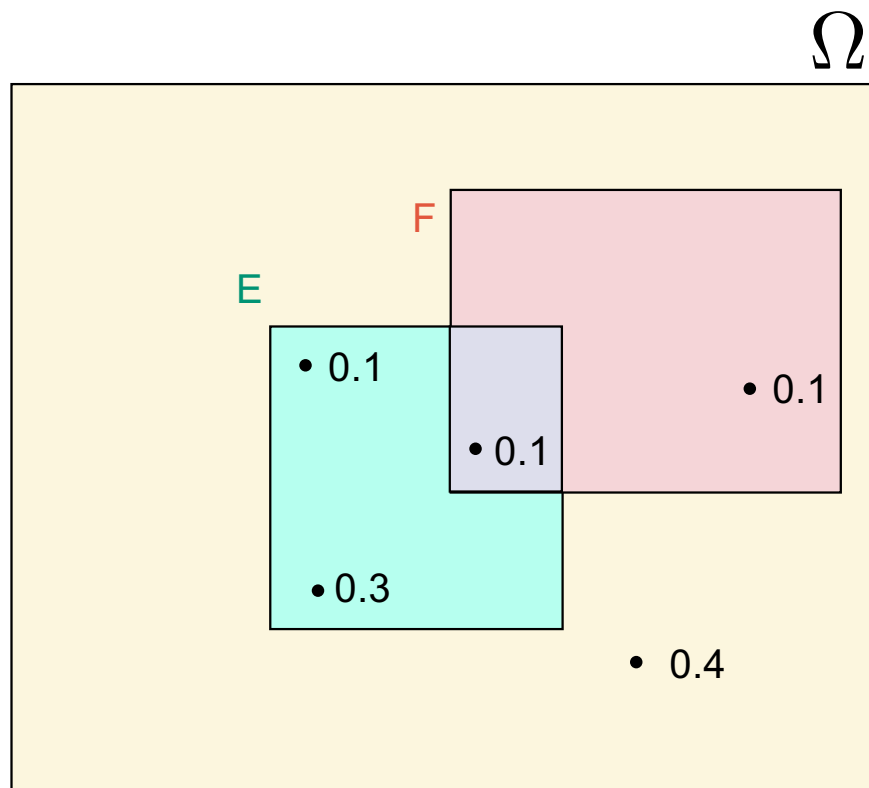
$$p(E|F) = \frac{p(E \cap F)}{p(F)}$$

Chain rule



$$p(E \cap F) = p(E|F) \times p(F)$$

Independence



$$P(E|F) = P(E)$$

$$p(E \cap F) = p(E) \times p(F)$$

Puzzle

When it rains, I forget my umbrella 5% of the time.

It rains every other day.

Last year, I had my umbrella on 271 days.

Puzzle

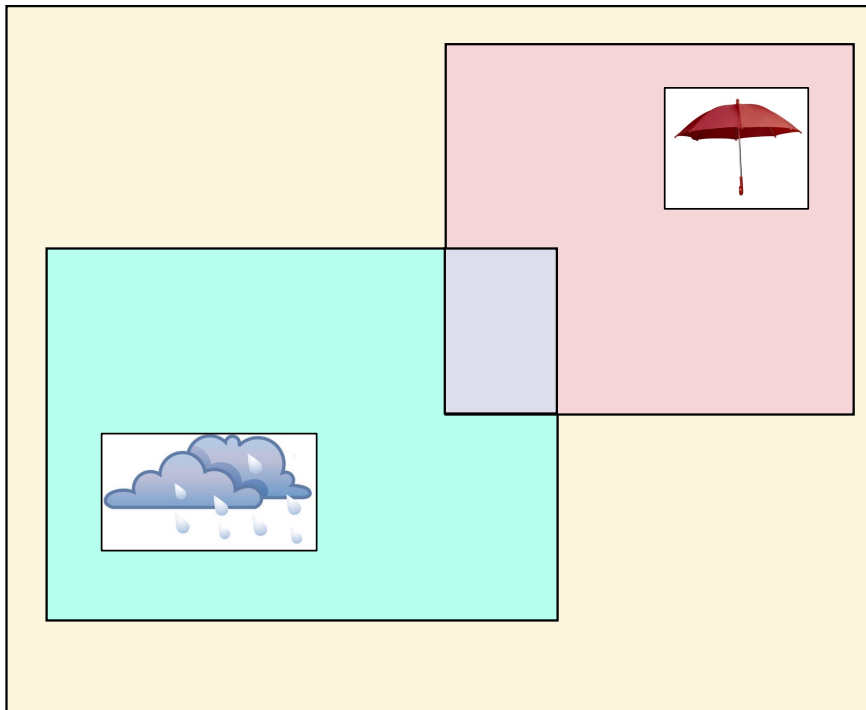
When it rains, I forget my umbrella 5% of the time.

It rains every other day.

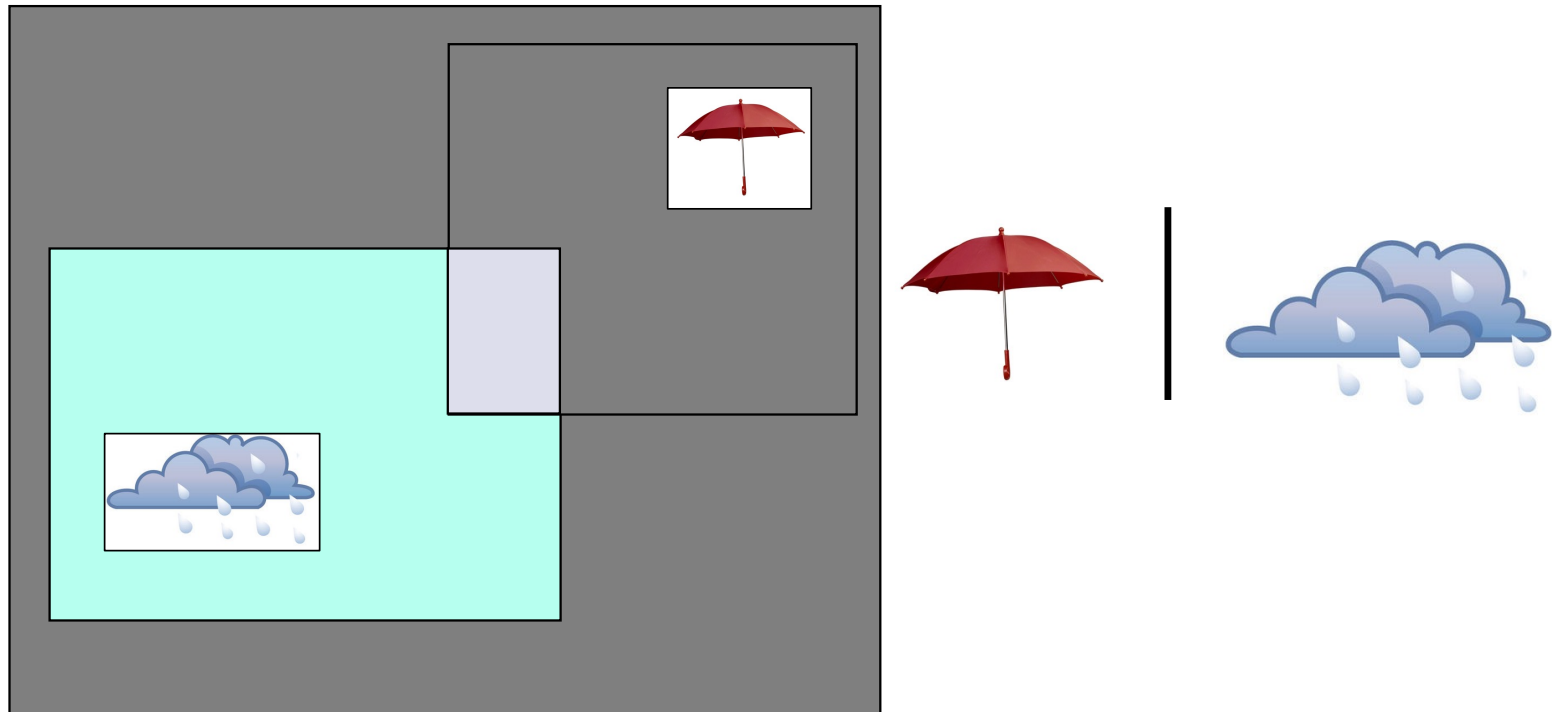
Last year, I had my umbrella on 271 days.

Today I have my umbrella. Is it raining?

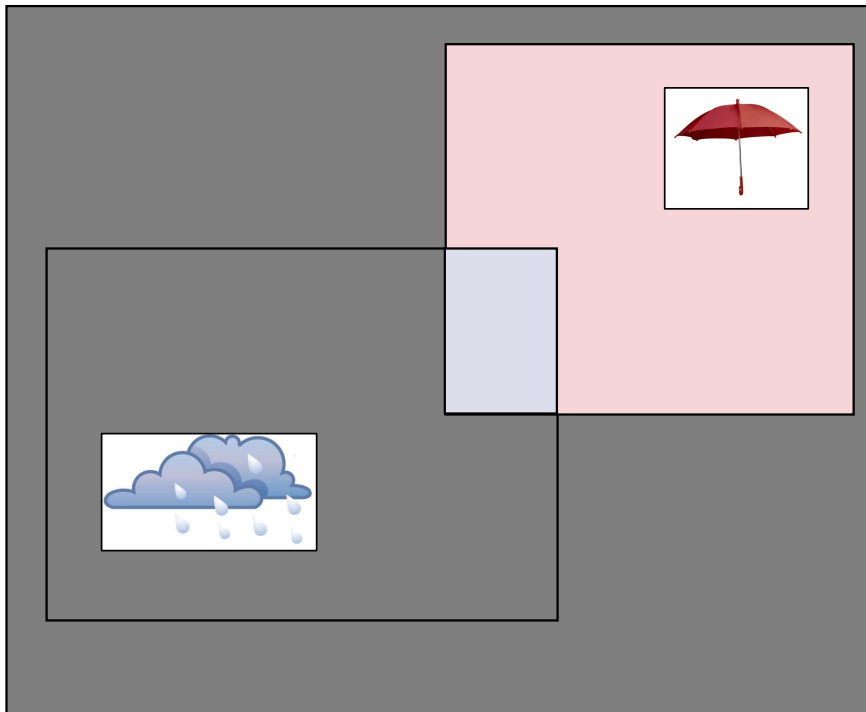
Visual representation



Visual representation



Visual representation



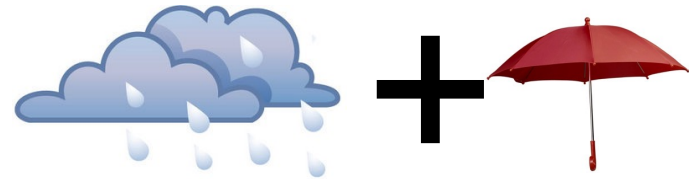
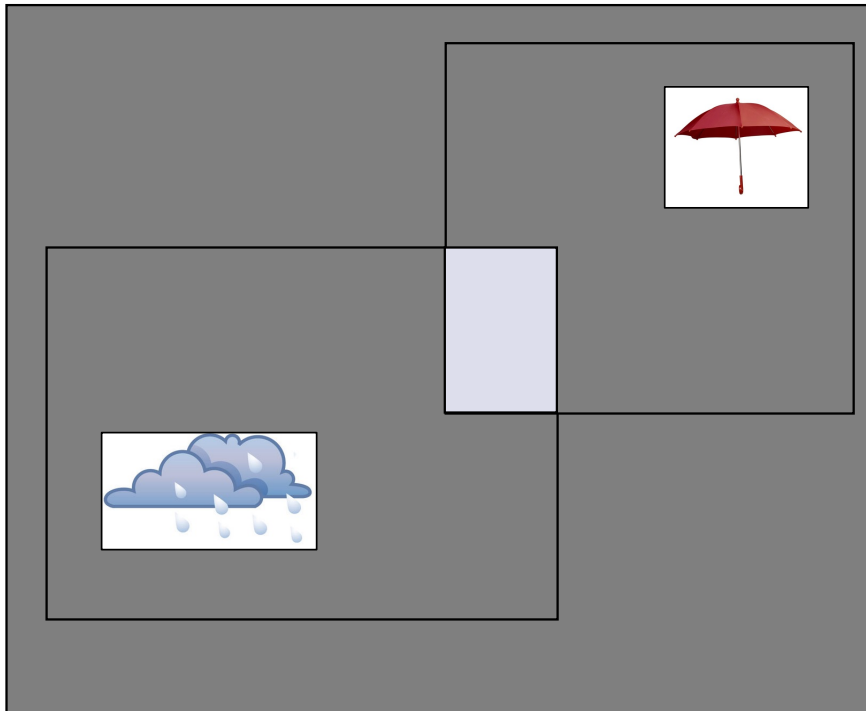
Probability that I don't forget my umbrella if it rains



Likelihood that it is raining if I took my umbrella



Visual representation



Bayes Formula – Using Chain Rule...

$P(\text{☂ and ☁})$

Bayes Formula – Using Chain Rule...

$$P(\text{☂ and ☁}) \\ = P(\text{☂} \mid \text{☁}).P(\text{☁})$$

Bayes Formula – Using Chain Rule...

$$P(\text{☂ and ☁})$$

$$= P(\text{☂} \mid \text{☁}).P(\text{☁})$$

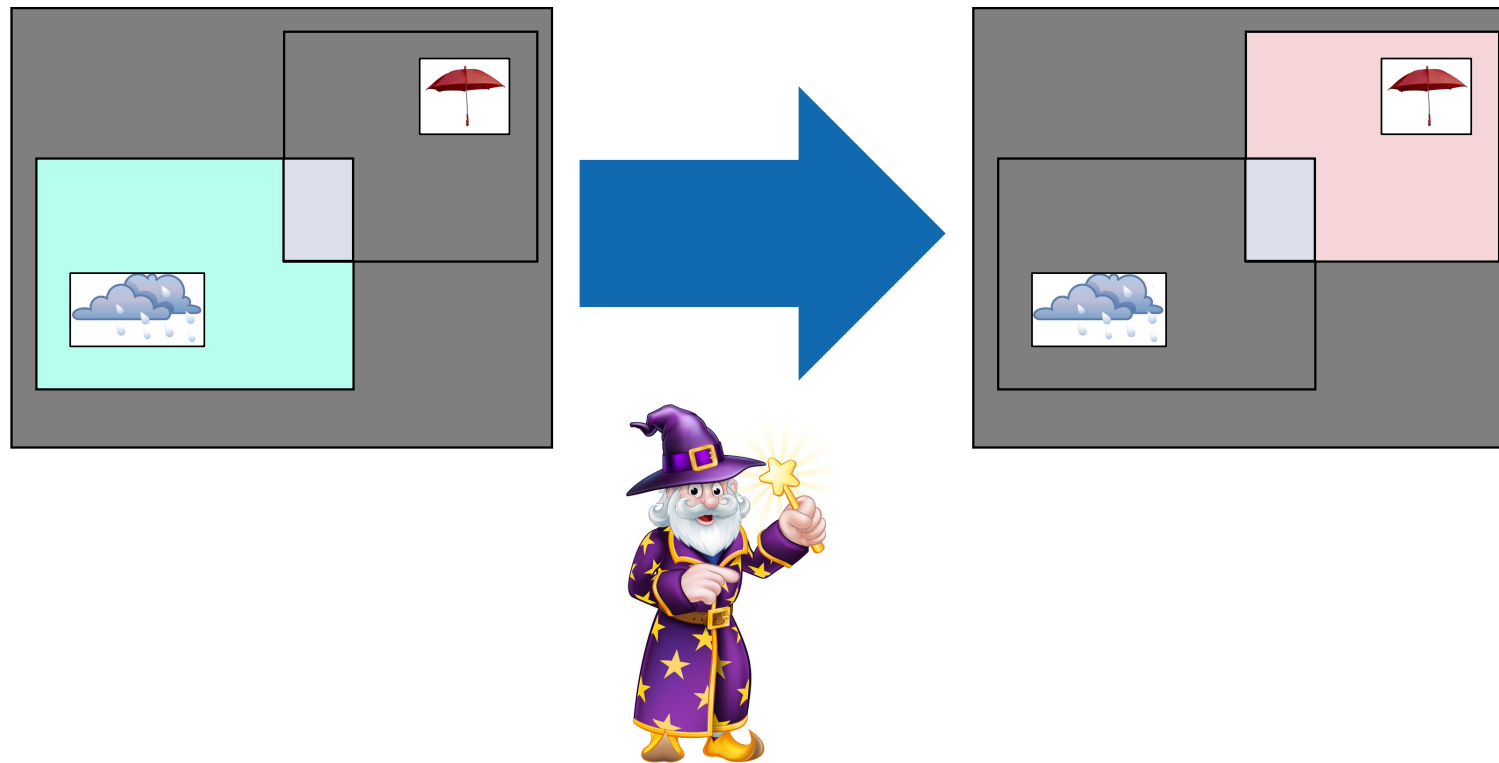
$$= P(\text{☁} \mid \text{☂}).P(\text{☂})$$

Bayes Formula

$$P(\text{cloud with rain} \mid \text{red umbrella})$$

$$= \frac{P(\text{red umbrella} \mid \text{cloud with rain}) \cdot P(\text{cloud with rain})}{P(\text{red umbrella})}$$

Visual representation



Bayes Formula

$$P(\text{☁️} \mid \text{☂️})$$

$$= \frac{0.95 \cdot P(\text{☁️})}{P(\text{☂️})}$$

Bayes Formula

$$P(\text{☁️} \mid \text{☂️})$$

$$= \frac{0.95 \cdot 0.5}{P(\text{☂️})}$$

Bayes Formula

$$P(\text{☁️} \mid \text{☂️})$$

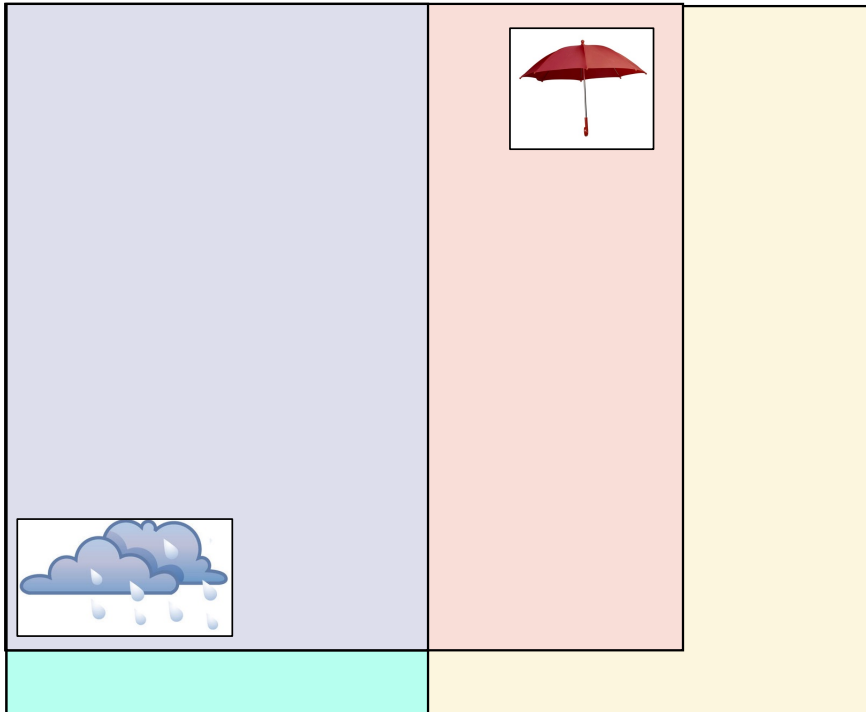
$$= \frac{0.95 \cdot 0.5}{271/365}$$

Bayes Formula

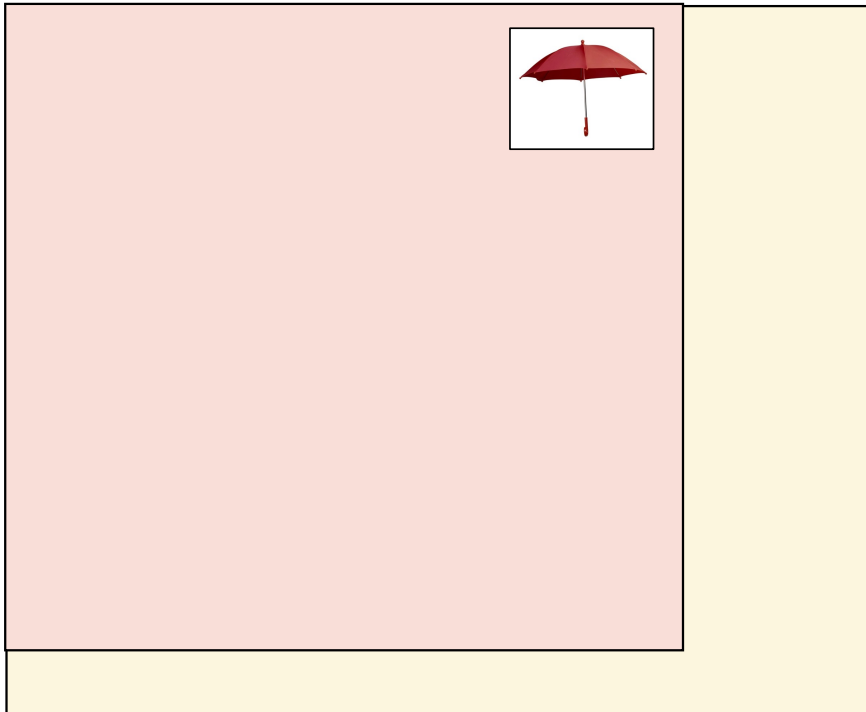
$$P(\text{☁️} \mid \text{☂️})$$

$$= 64\%$$

Visual representation (at scale)

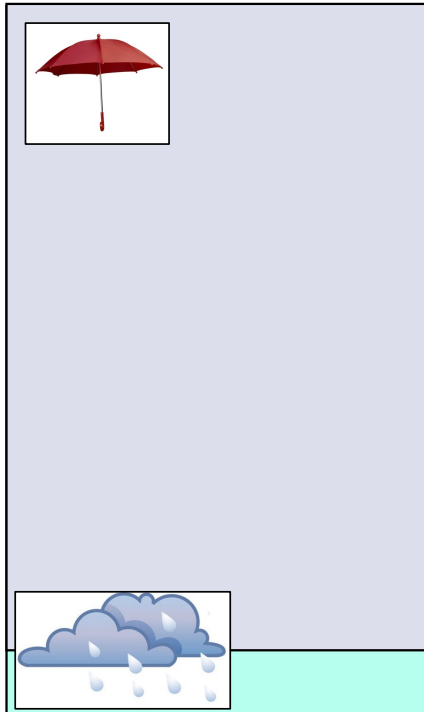


Visual representation (at scale)



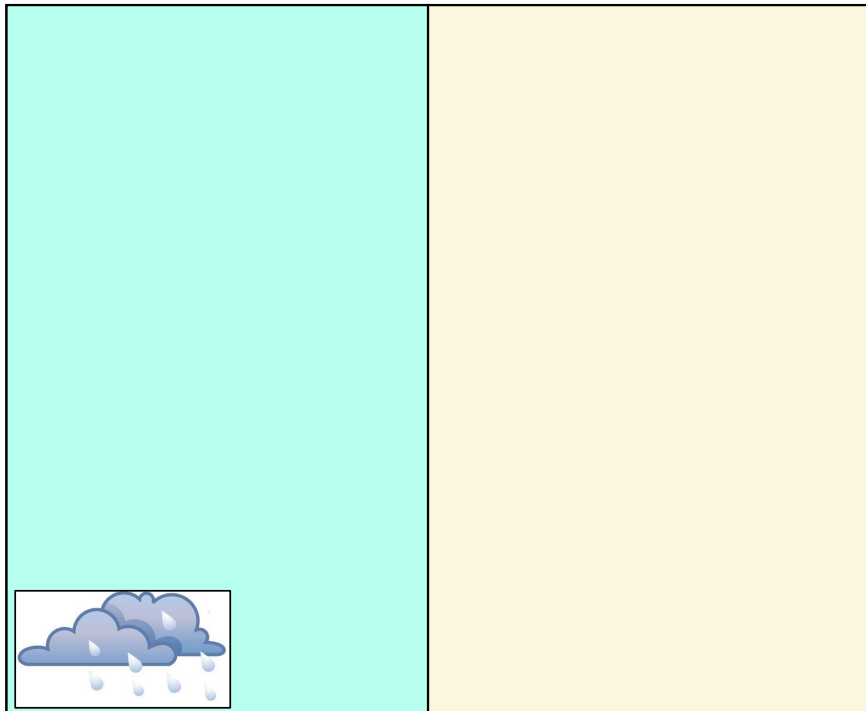
I had my umbrella 271 of 365 days.

Visual representation (at scale)



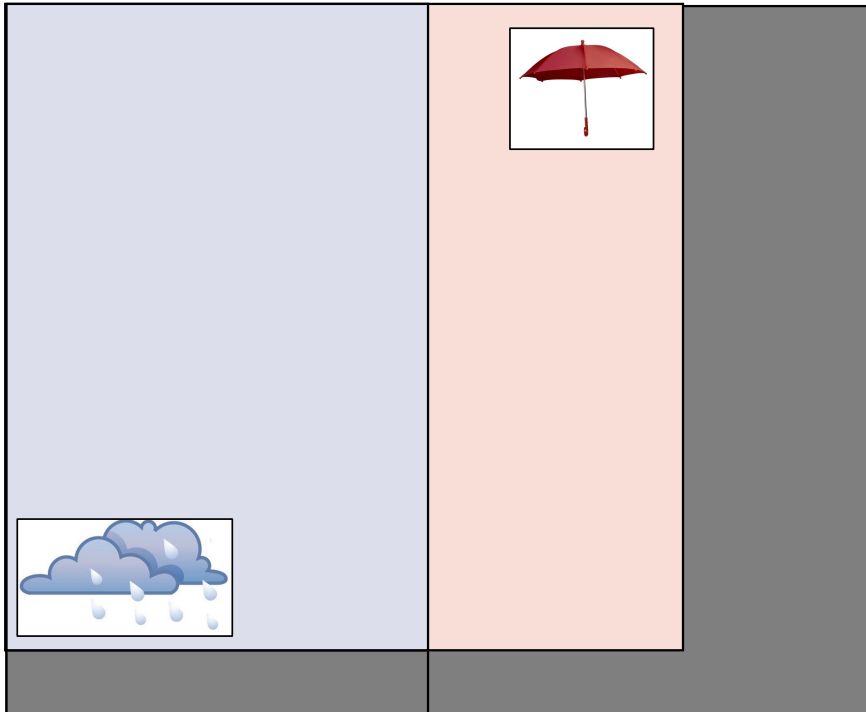
I only forget my umbrella 5% of the rainy days.

Visual representation (at scale)



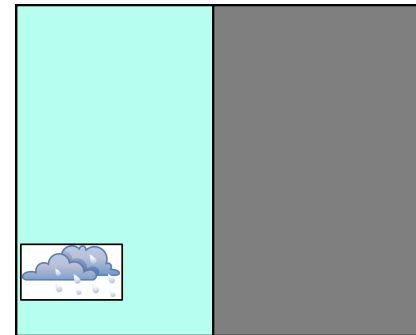
It rains 50% of the time

Visual representation (at scale)



But it rains 64% of the time
on the days I have my umbrella.

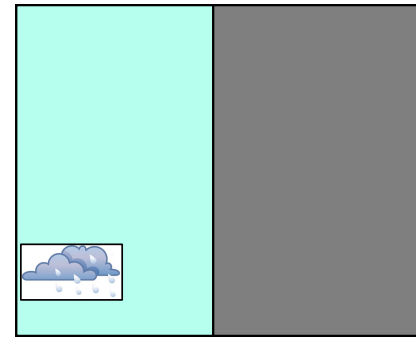
The magics



P()

prior

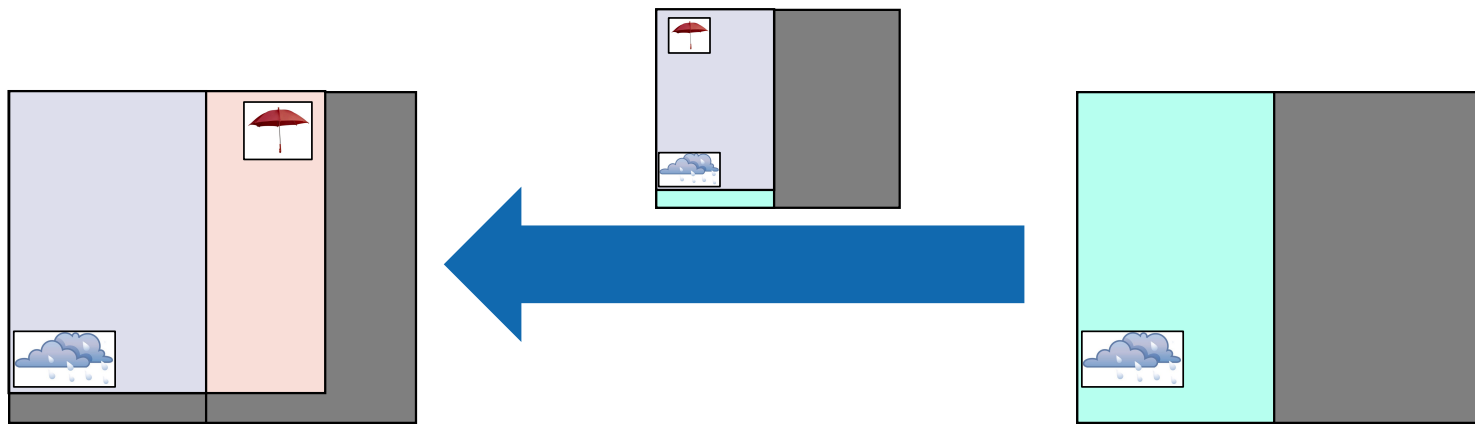
Prior



P()

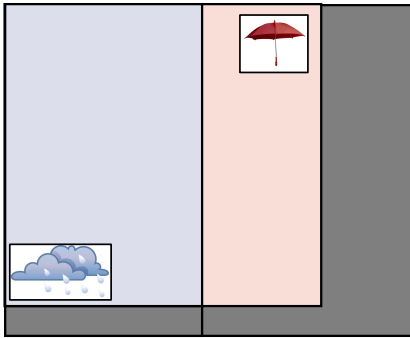
prior

Adding new information



$$P(\text{cloud with rain} \mid \text{red umbrella}) = \frac{P(\text{red umbrella} \mid \text{cloud with rain})}{P(\text{red umbrella})} \times P(\text{cloud with rain})$$

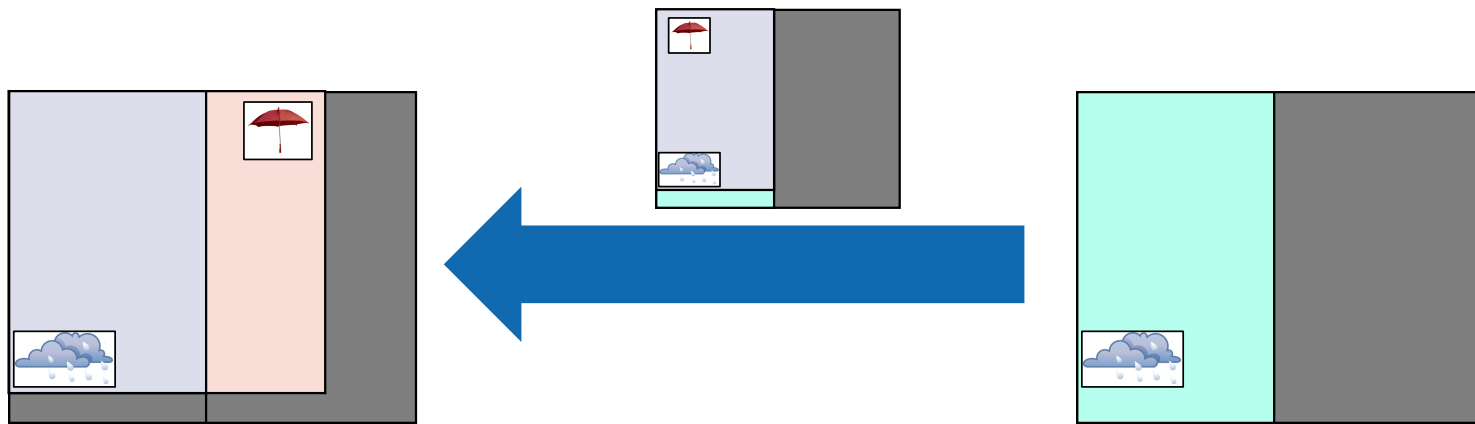
Posterior



$$P(\text{cloud with rain} \mid \text{umbrella}) =$$

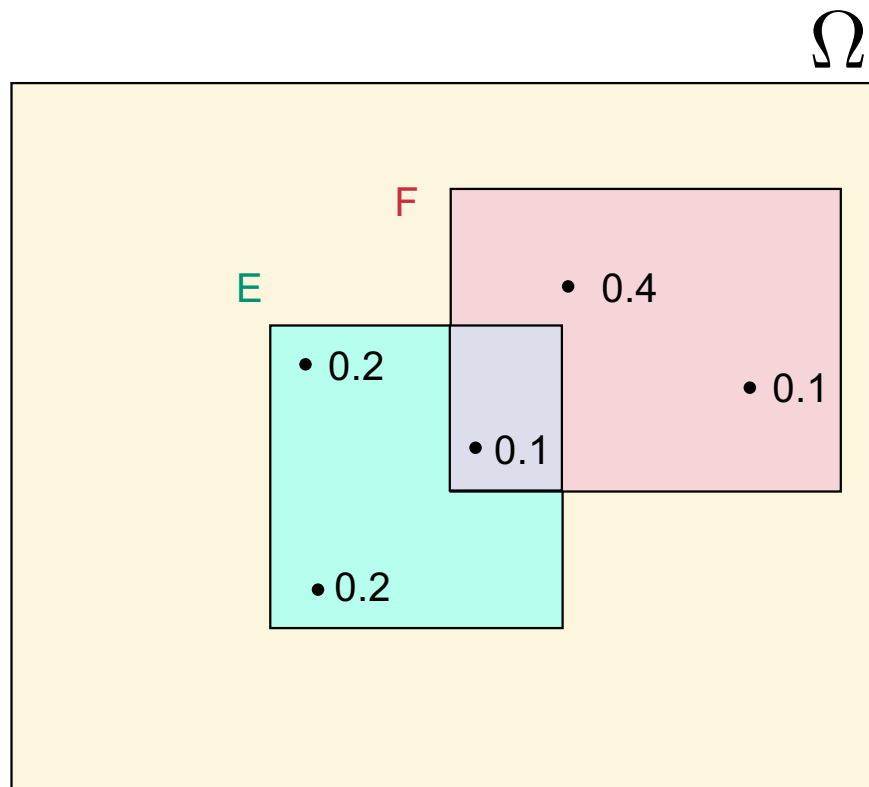
posterior

The magics



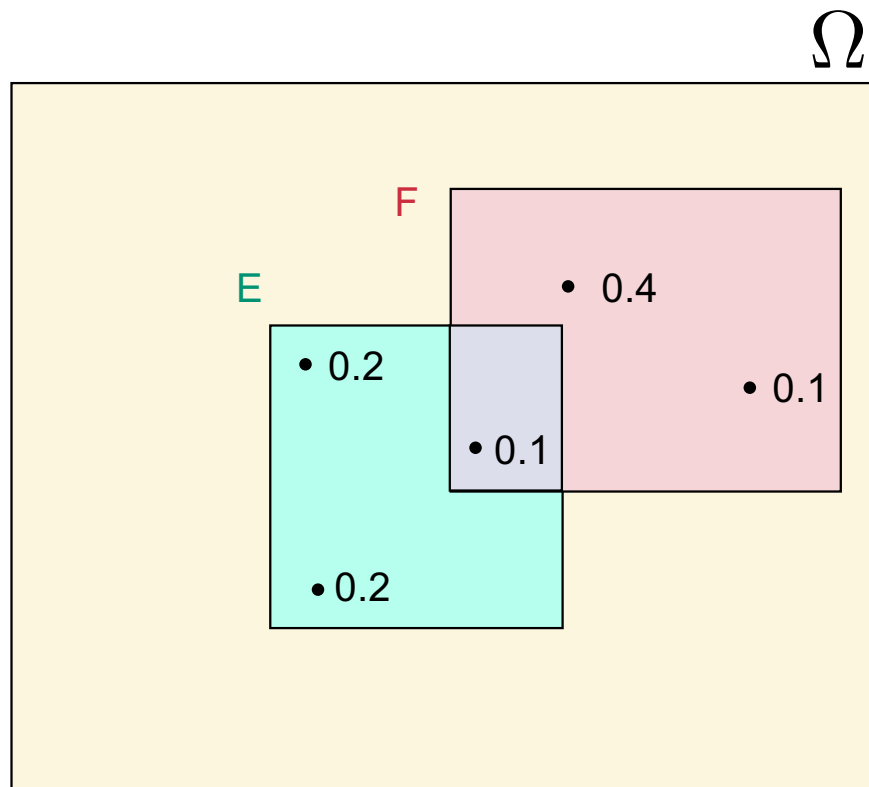
$$\underbrace{P(\text{cloud} \mid \text{umbrella})}_{\text{posterior}} = \frac{P(\text{umbrella} \mid \text{cloud})}{P(\text{umbrella})} \times \underbrace{P(\text{cloud})}_{\text{prior}}$$

Bayes' rule



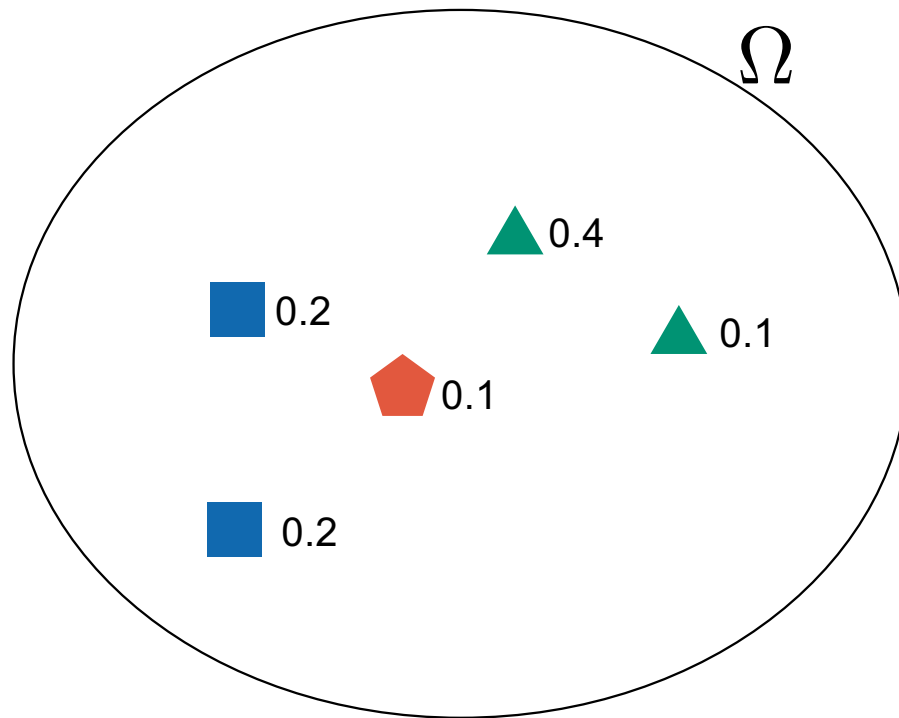
$$p(E|F) = \frac{P(F|E) \times P(E)}{P(F)}$$

Bayes' rule



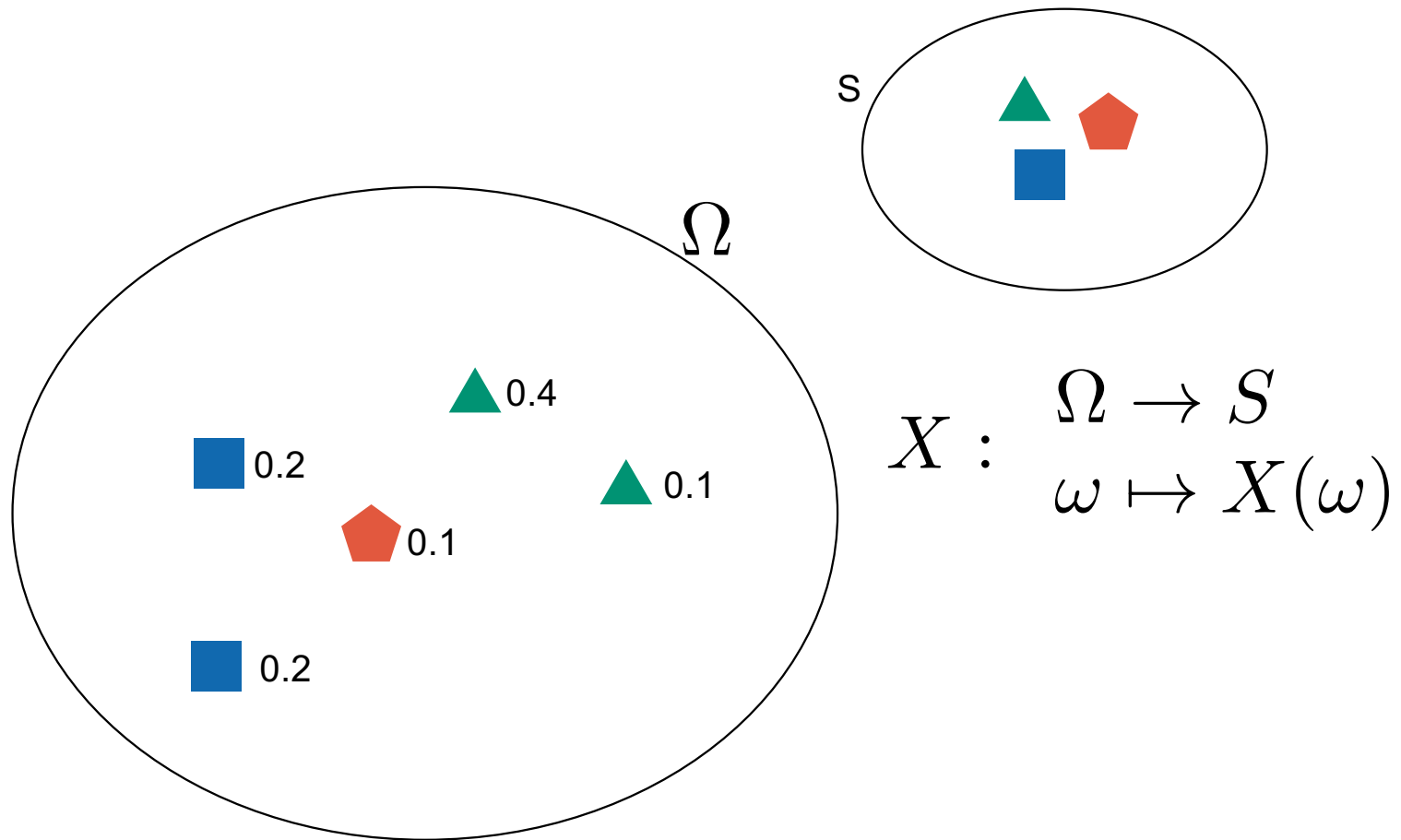
$$\underbrace{p(E|F)}_{\text{posterior}} = \frac{P(F|E)}{P(F)} \times \underbrace{P(E)}_{\text{prior}}$$

Random variable

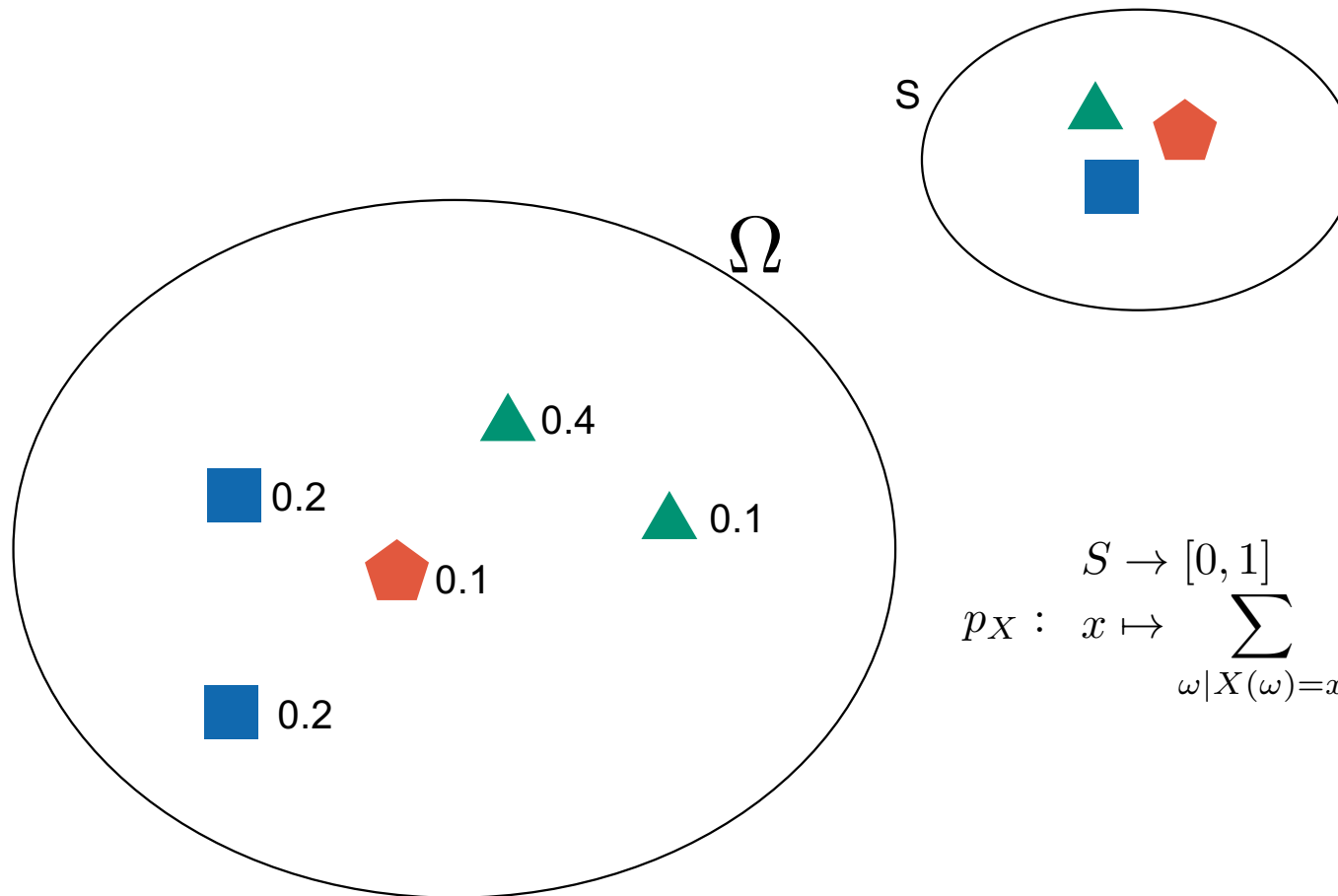


$$X : \begin{array}{l} \Omega \rightarrow S \\ \omega \mapsto X(\omega) \end{array}$$

Random variable

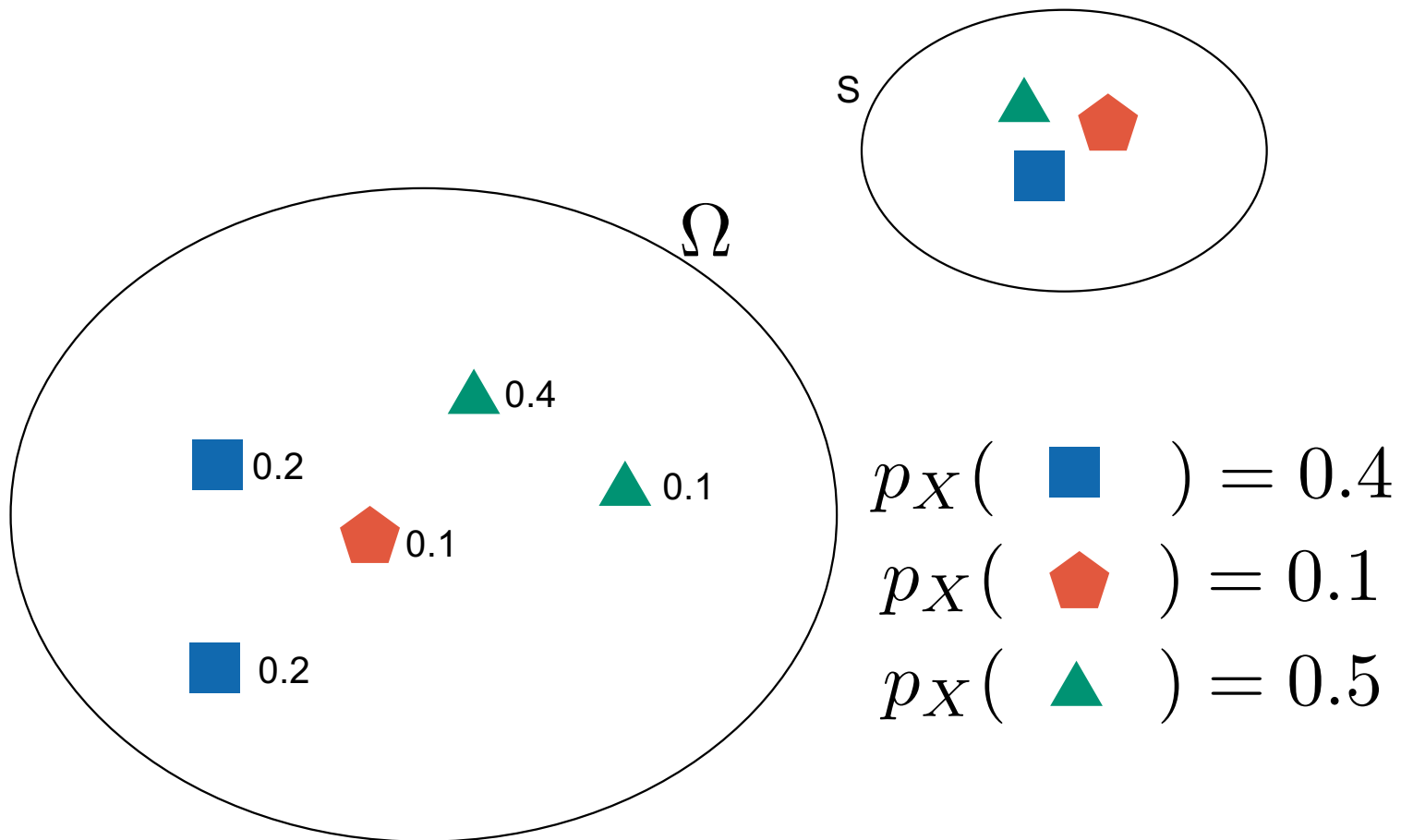


Random variable

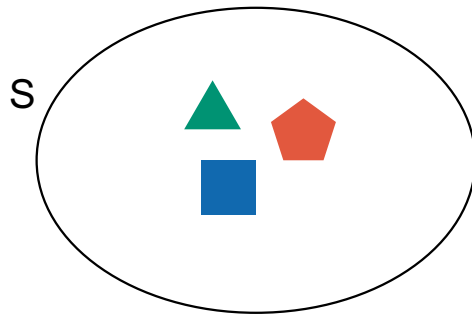


$$p_X : \begin{matrix} S \rightarrow [0, 1] \\ x \mapsto \sum_{\omega | X(\omega)=x} p(\omega) \end{matrix}$$

Random variable



In practice...

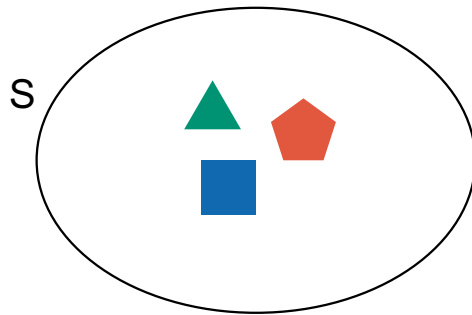


$$p_X(\text{blue square}) = 0.4$$

$$p_X(\text{red pentagon}) = 0.1$$

$$p_X(\text{green triangle}) = 0.5$$

Alternate notation

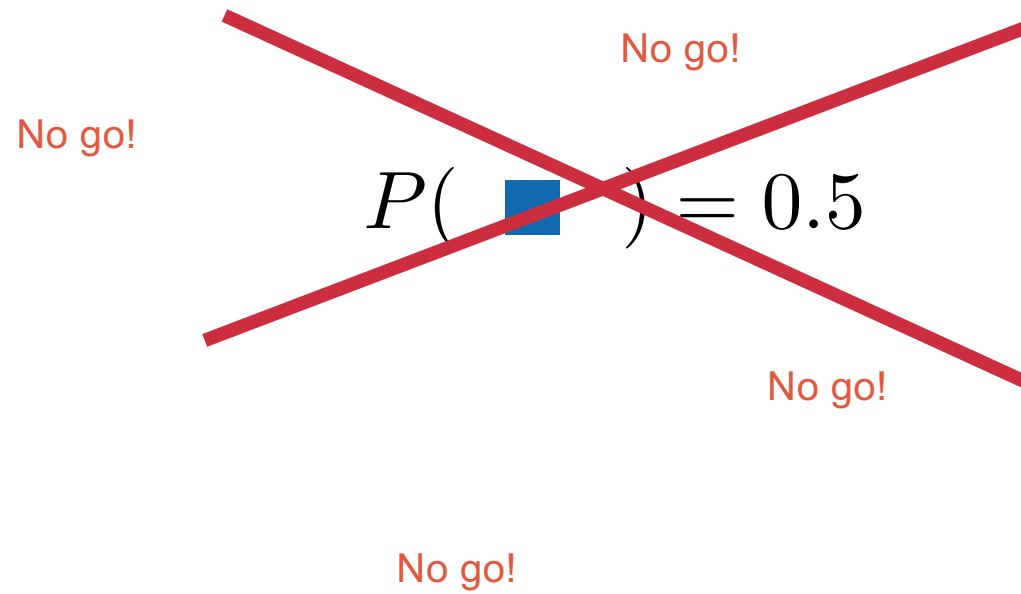


$$P(X = \text{blue square}) = 0.4$$

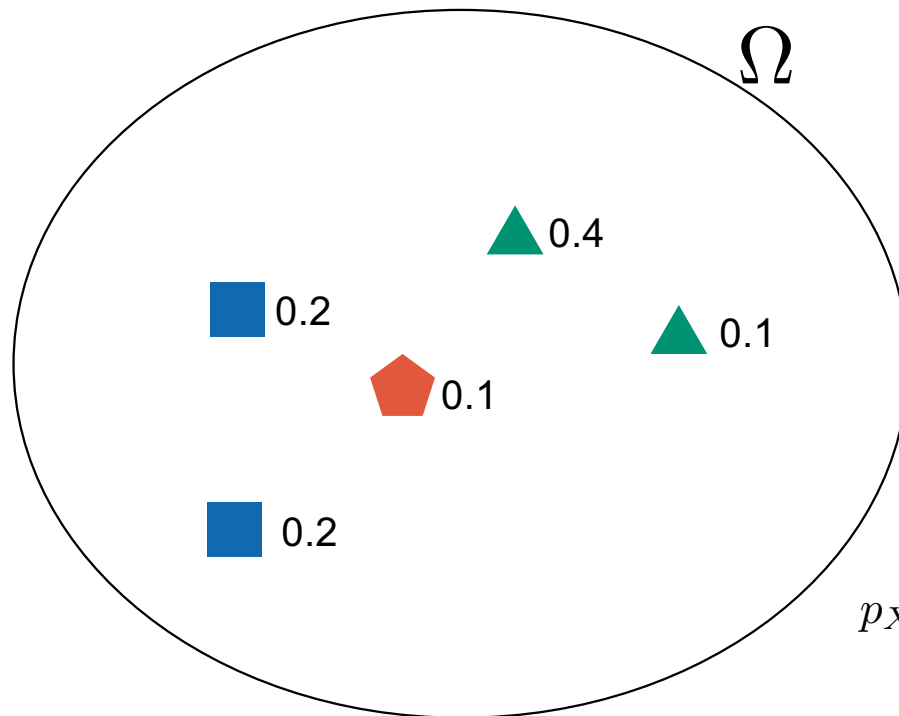
$$P(X = \text{red pentagon}) = 0.1$$

$$P(X = \text{green triangle}) = 0.5$$

Don't do that!

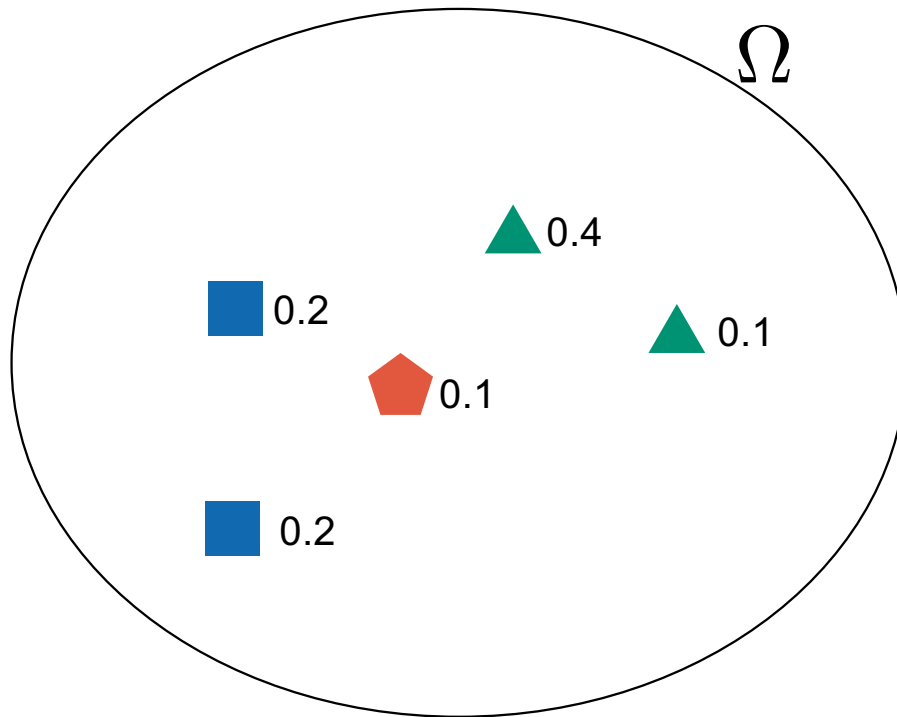


Joint probabilities



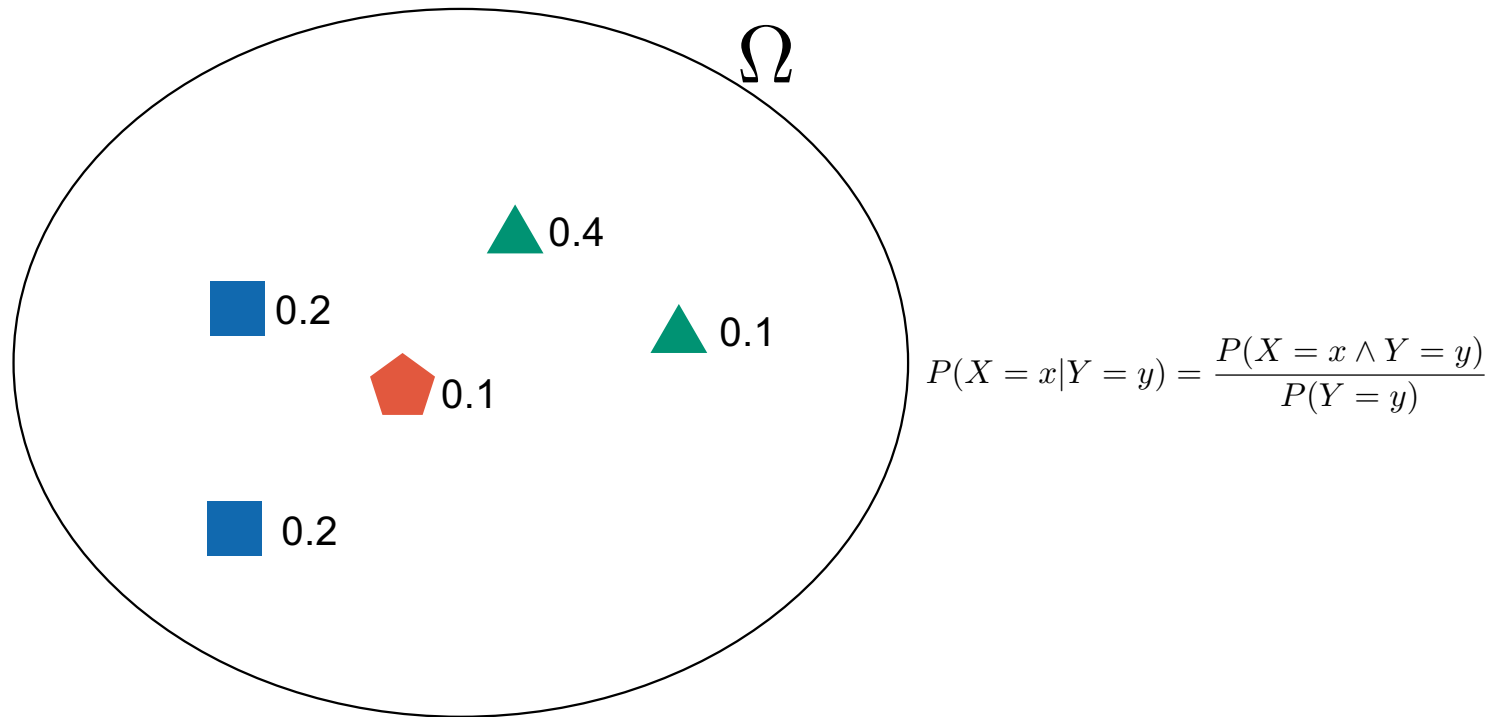
$$p_{XY}(x, y) = \sum_{\omega | X(\omega)=x \wedge Y(\omega)=y} p(\omega)$$

Conditional probabilities

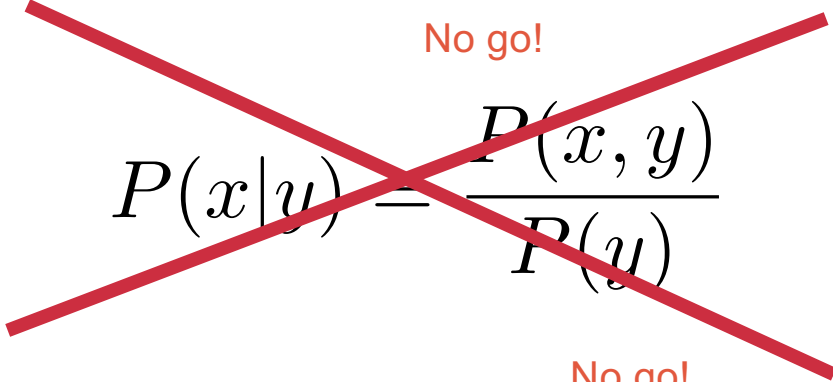


$$p_{X|Y}(x, y) = \frac{p_{XY}(x, y)}{p_Y(y)}$$

Conditional probabilities



Don't do that!

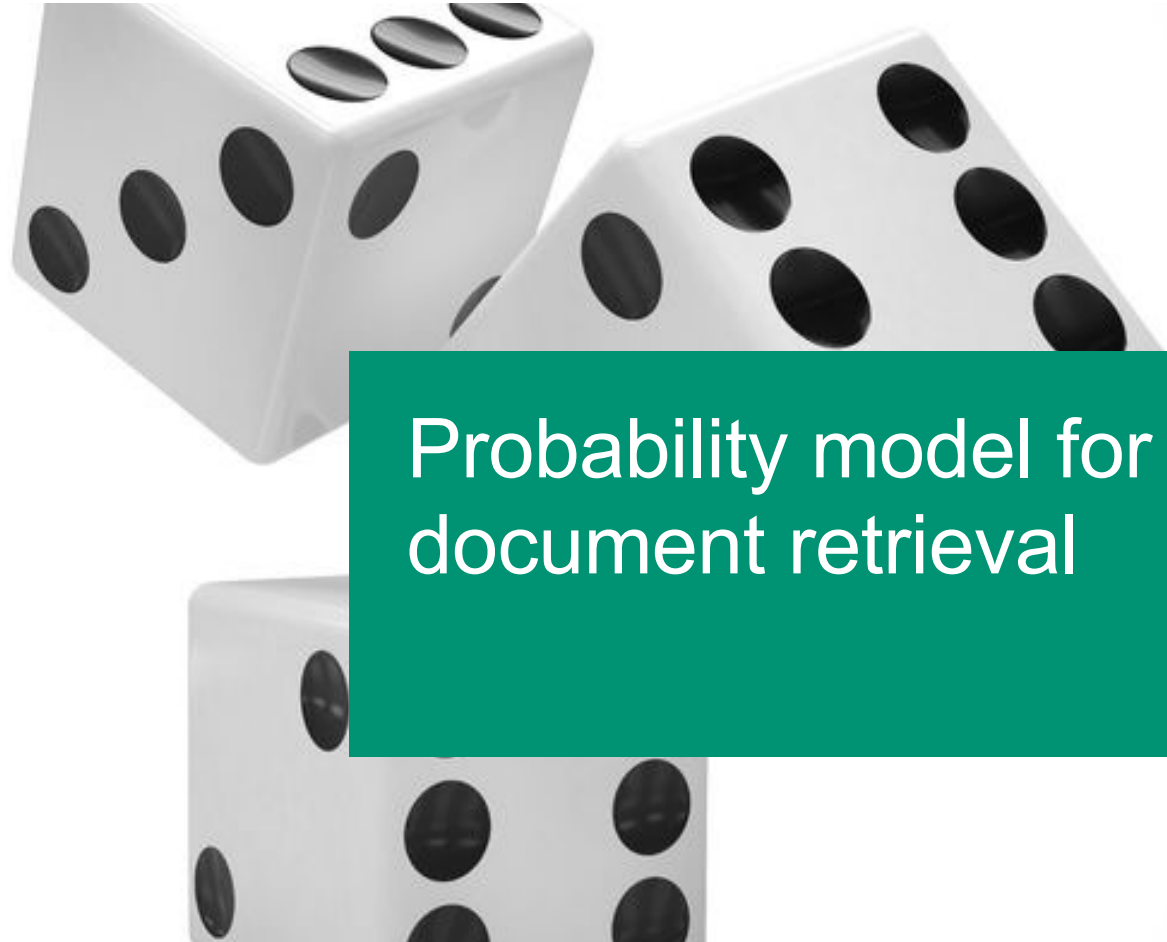


No go!

$$P(x|y) = \frac{P(x, y)}{P(y)}$$

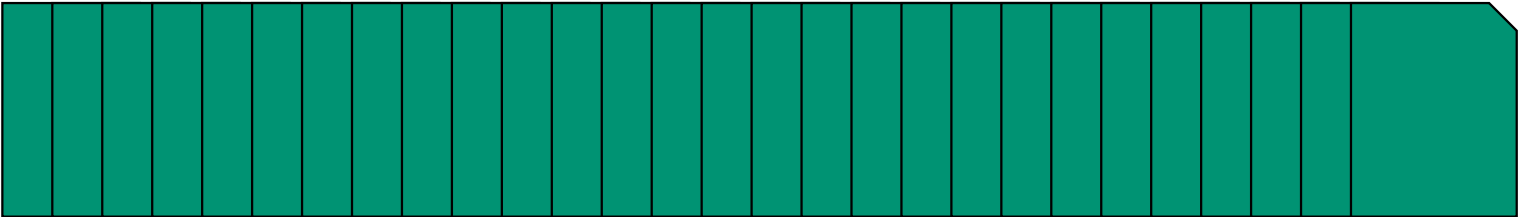
No go!

No go!



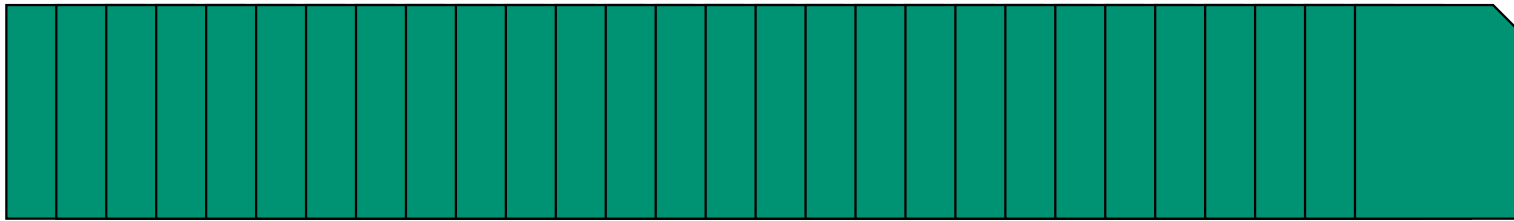
Probability model for
document retrieval

Evaluation setup

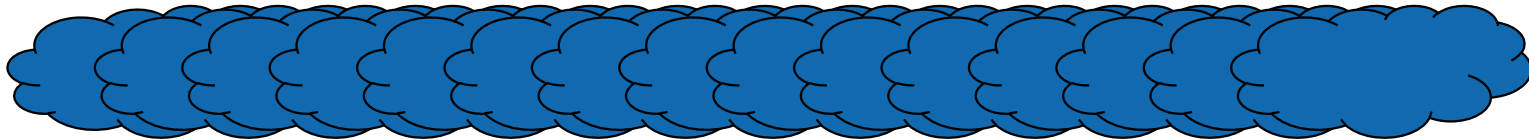


Collection of documents

Evaluation setup

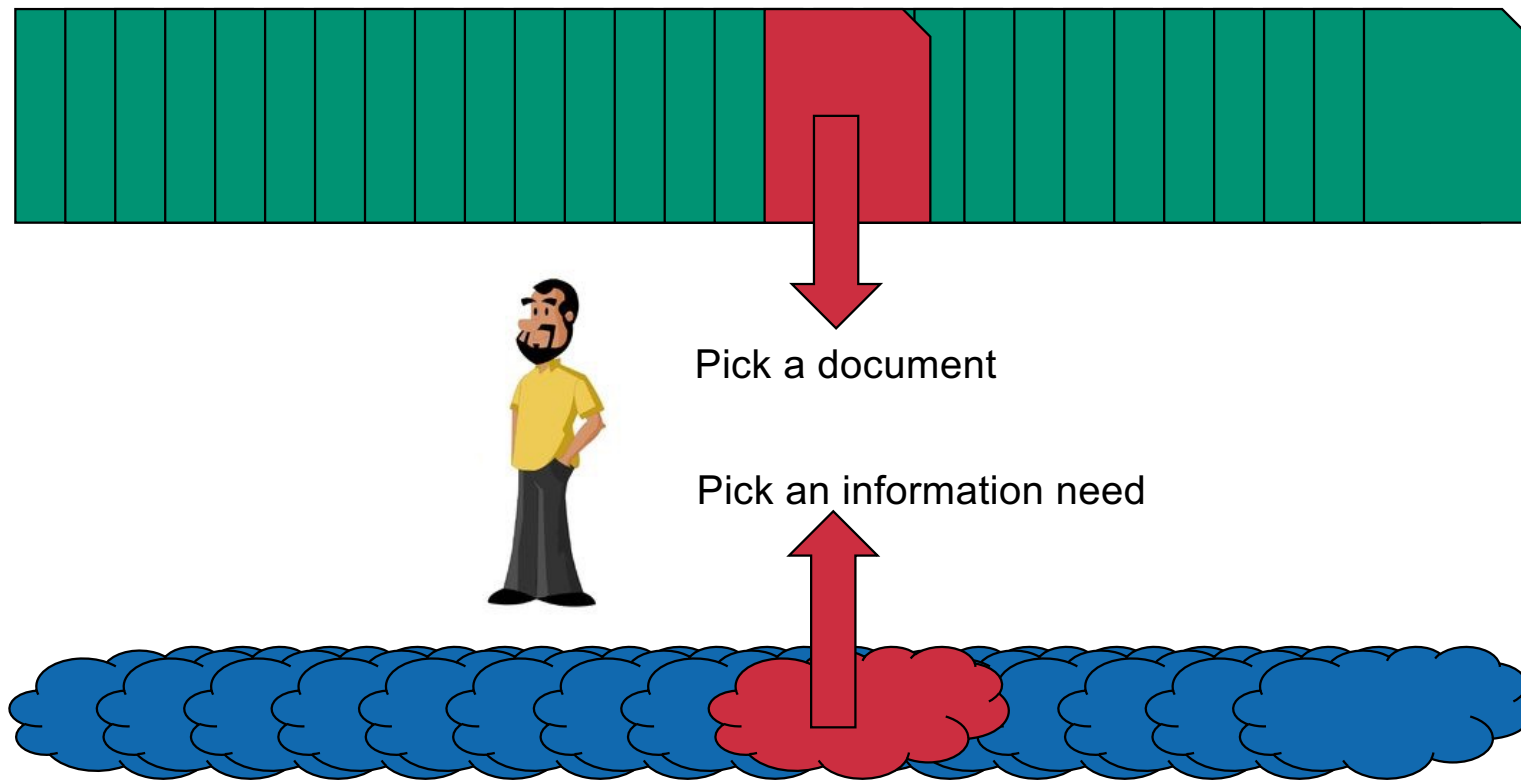


Collection of documents

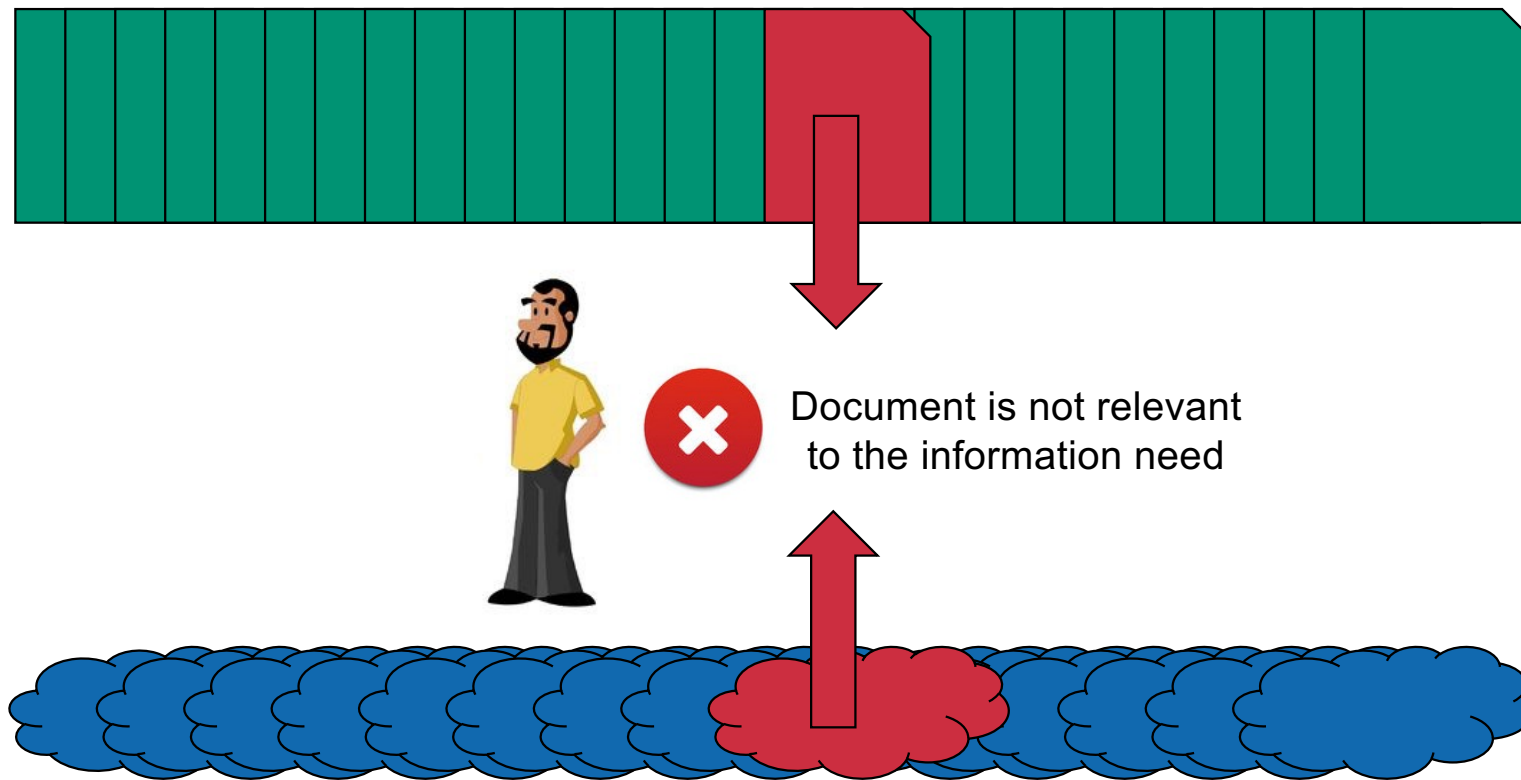


Set of information needs

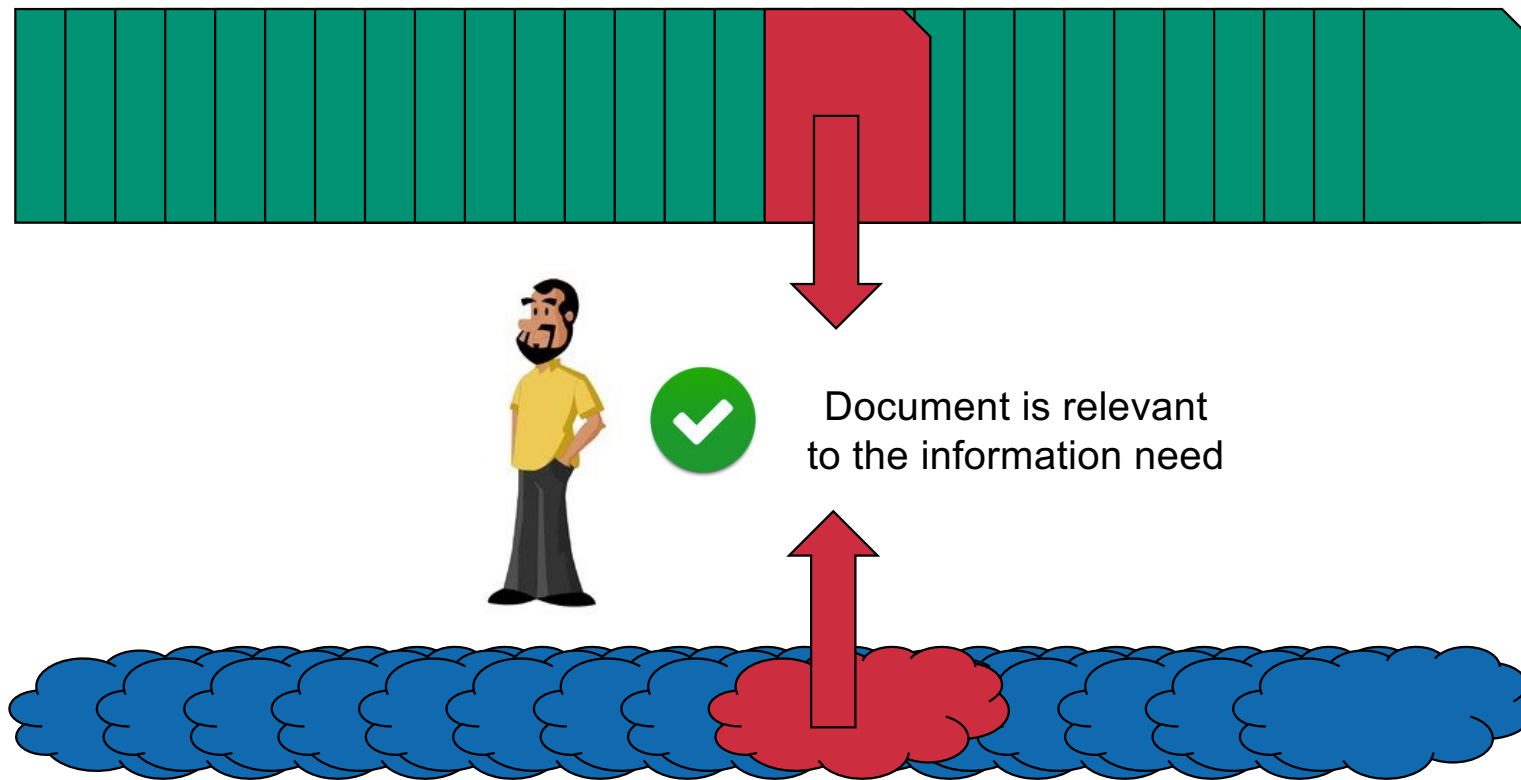
Evaluation setup



Evaluation setup



Evaluation setup



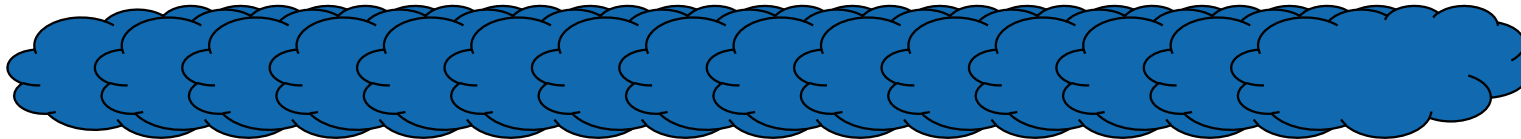
Cranfield dataset



1'398 documents

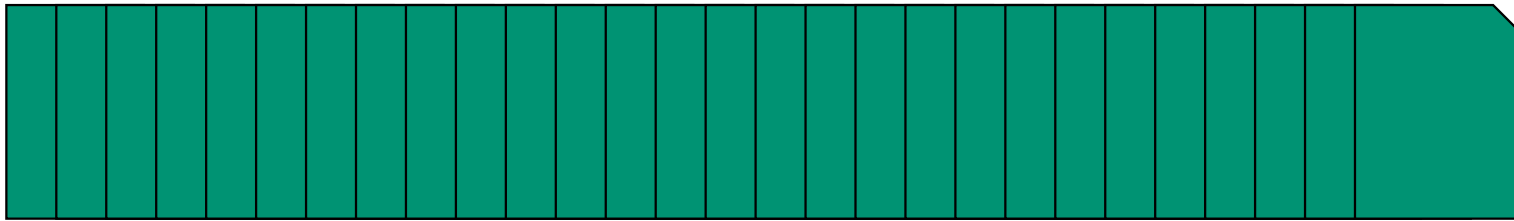


Relevance judgement for each combination



225 information needs

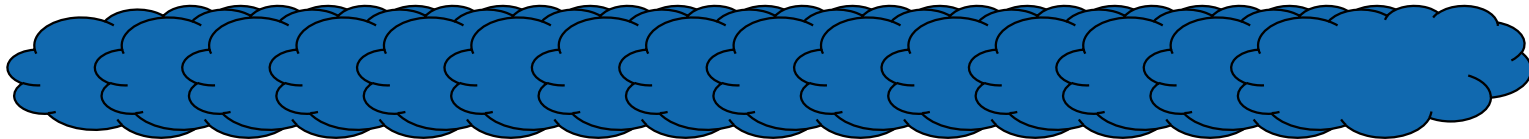
TREC dataset



1,890,000 documents



Relevance judgement for each combination



450 information needs

TREC dataset



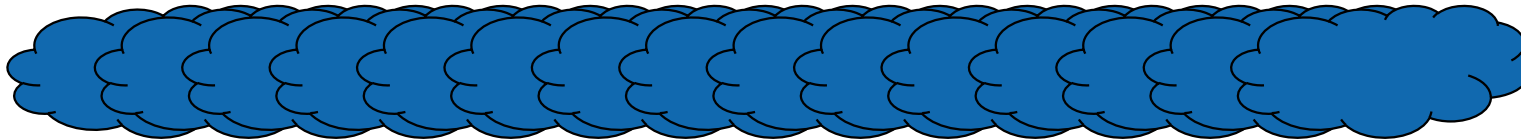
1,890,000 documents



Domain experts

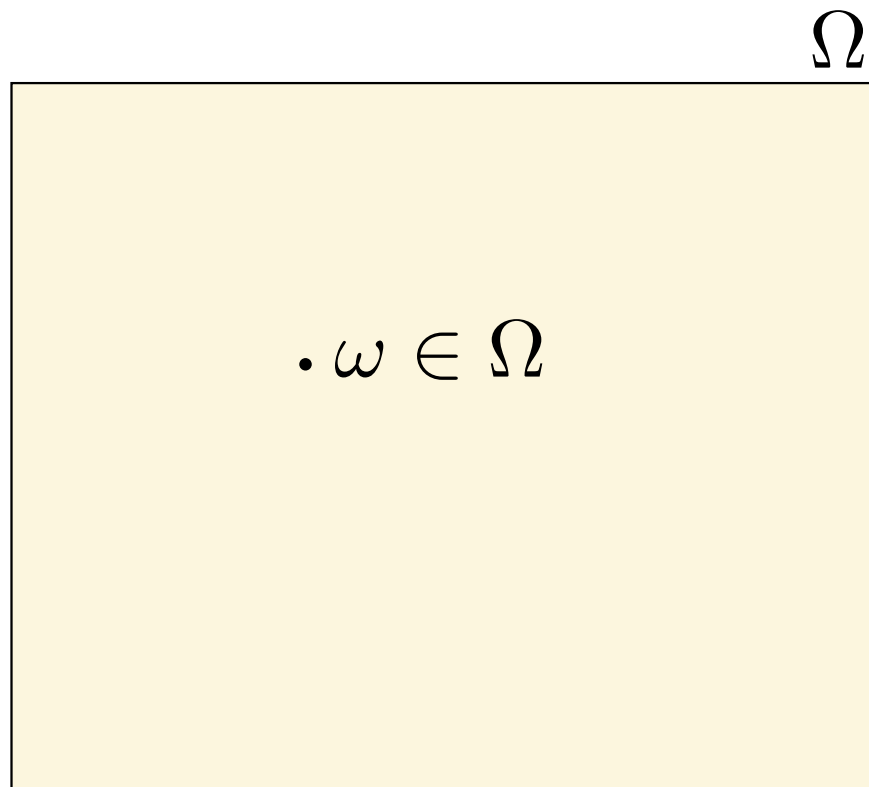


Relevance judgement for each combination



450 information needs

Elementary event (outcome)



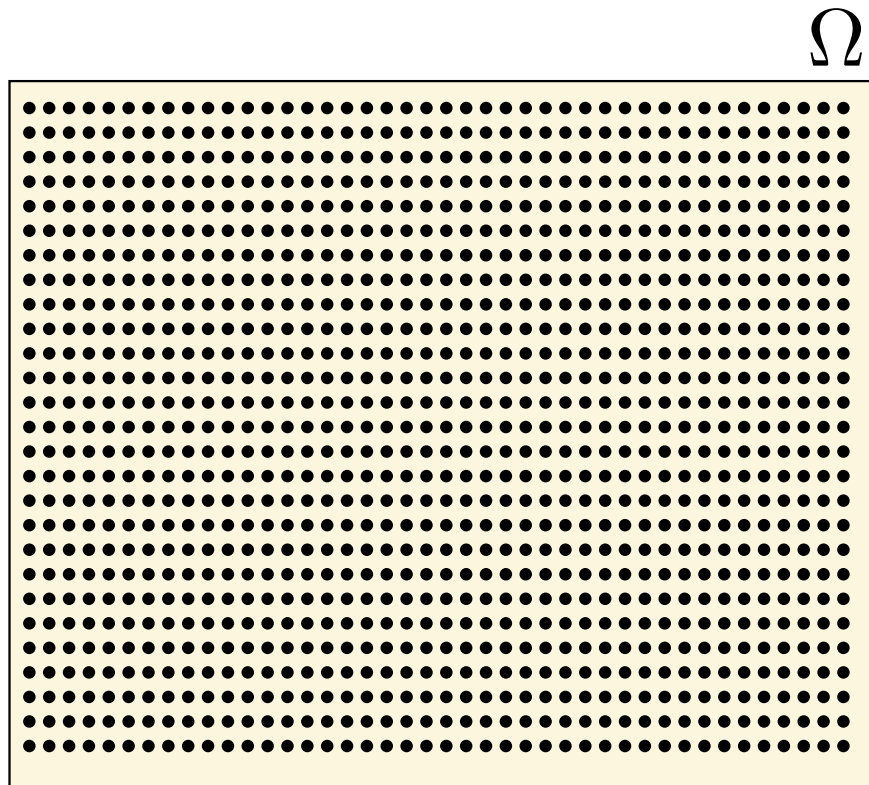
Elementary event

=

We gave a domain expert
one **query**
and
one **document**

The domain expert said
whether **yes** or **no**, the
document
is **relevant** to the
query

Elementary event (outcome)



Ideal setup:

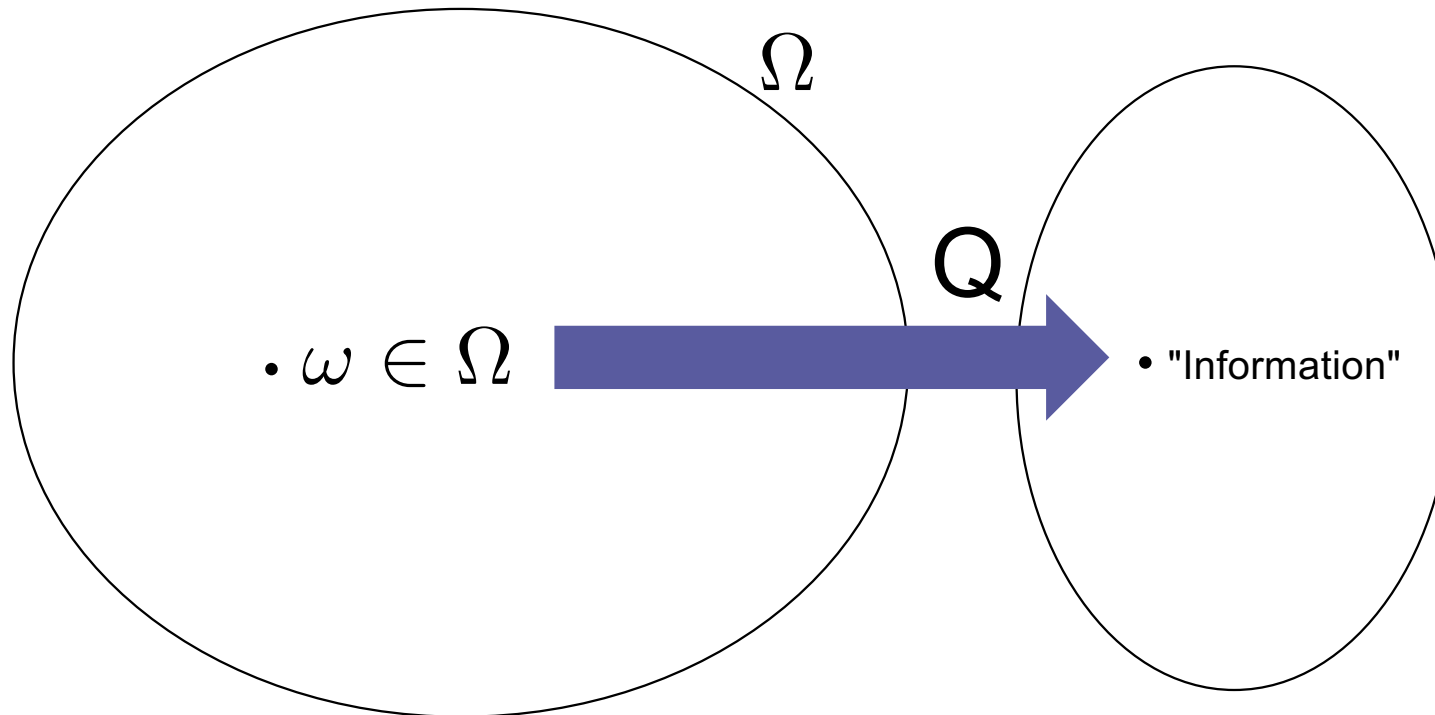
lots of queries

lots of documents

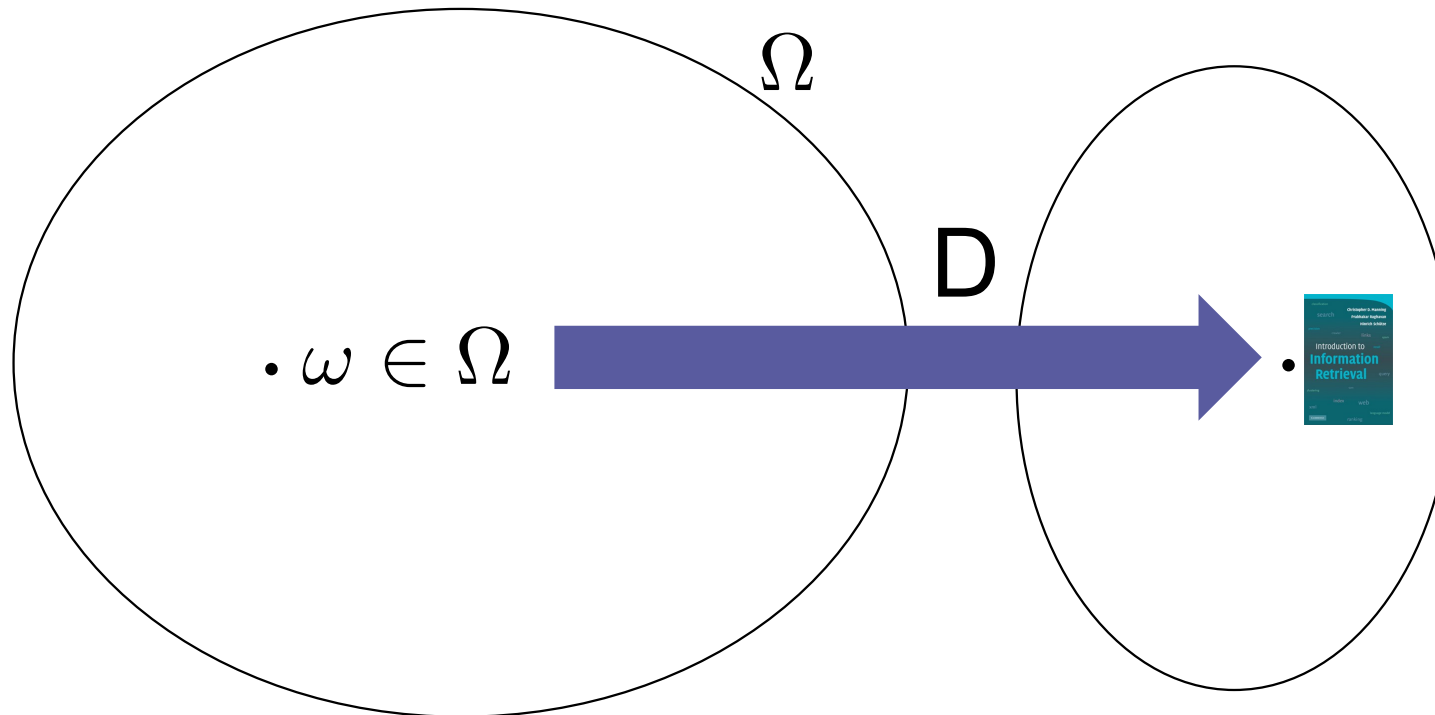
lots of domain experts

Thought experiment!

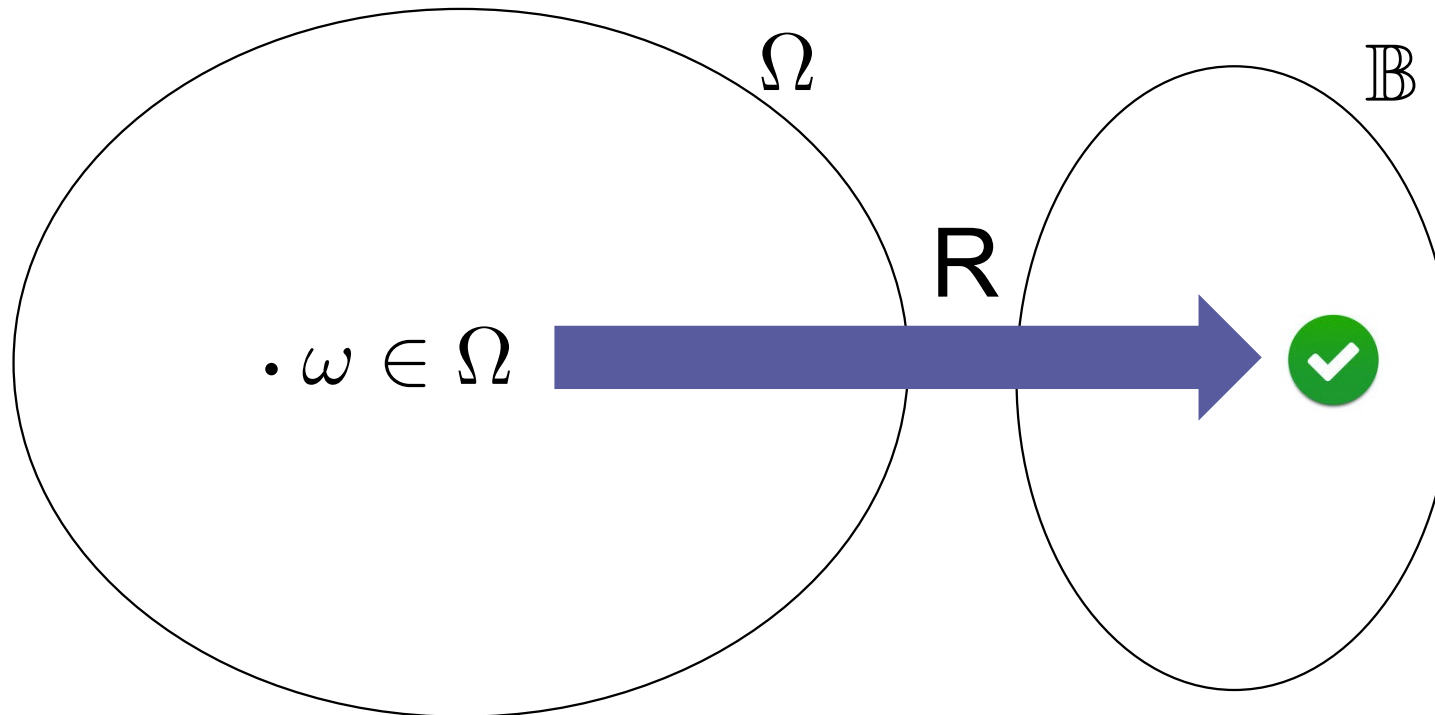
Query as a random variable



Document as a random variable



Relevance as a random variable

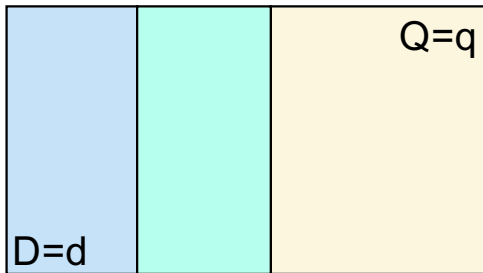


Probability ranking principle

$$P(R = 1 | D = d \wedge Q = q)$$

Probability that, for a query q and a document d ,
 d is relevant for query q

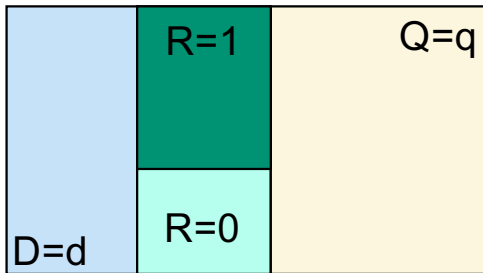
Probability ranking principle



$$P(R = 1 | D = d \wedge Q = q)$$

Probability that, for a query q and a document d ,
 d is relevant for query q

Probability ranking principle



$$P(R = 1 | D = d \wedge Q = q)$$

Probability that, for a query q and a document d ,
 d is relevant for query q

Ideal world

$$P(R = 1 | D = d \wedge Q = q)$$


$$P(R = 1 | D = e \wedge Q = q)$$


$$P(R = 1 | D = f \wedge Q = q)$$


$$P(R = 1 | D = g \wedge Q = q)$$


Probability ranking principle

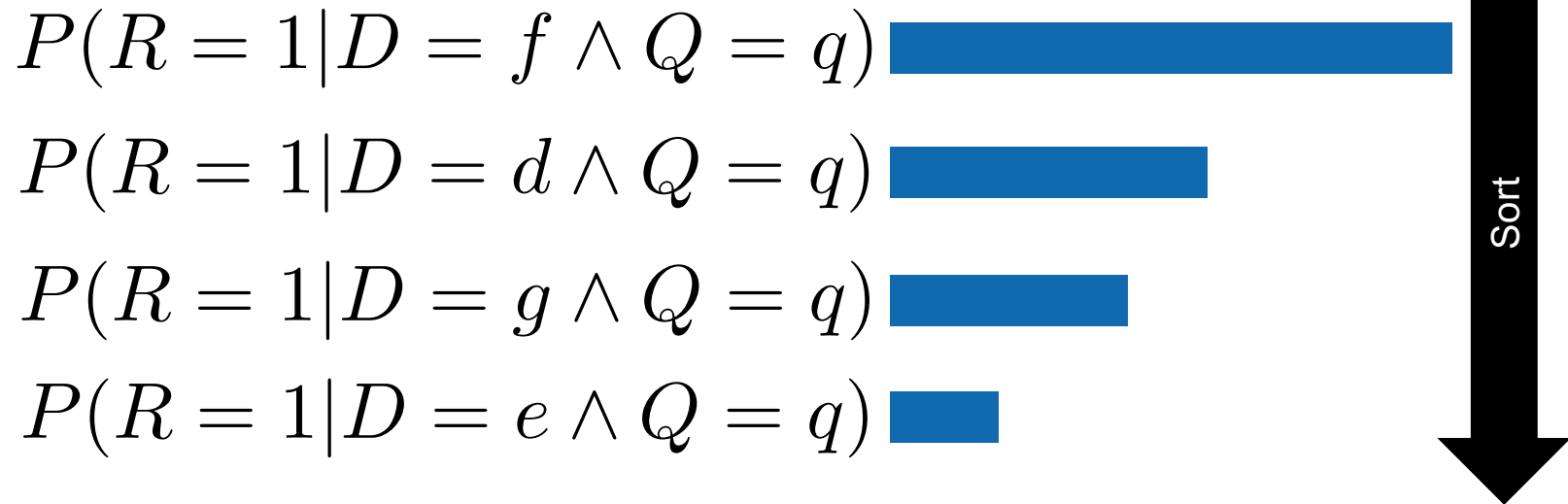
$$P(R = 1 | D = d \wedge Q = q) \text{ } \text{■}$$

$$P(R = 1 | D = e \wedge Q = q) \text{ } \text{■}$$

$$P(R = 1 | D = f \wedge Q = q) \text{ } \text{■}$$

$$P(R = 1 | D = g \wedge Q = q) \text{ } \text{■}$$

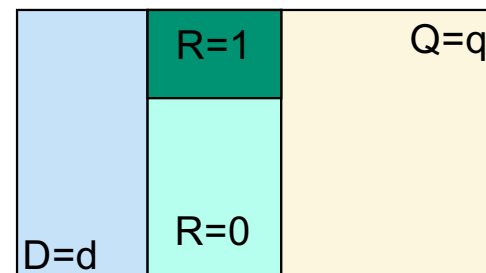
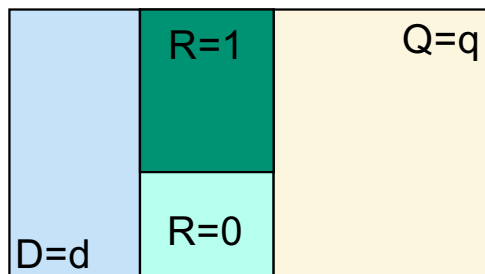
Probability ranking principle



"Boolean retrieval"

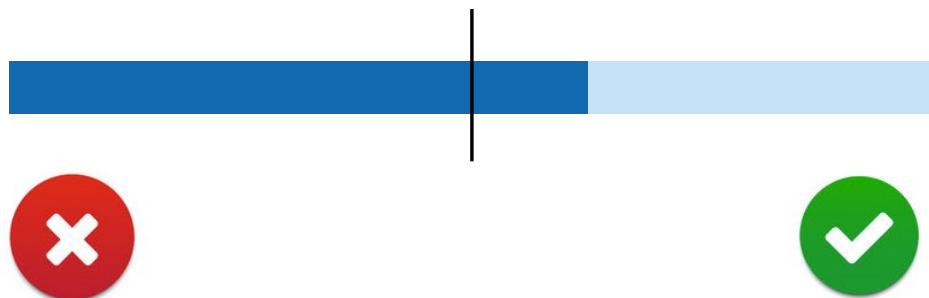
$$P(R = 1|D = d \wedge Q = q) > P(R = 0|D = d \wedge Q = q)$$

"Boolean retrieval"



"Boolean retrieval"

$$P(R = 1 | D = d \wedge Q = q) > \frac{1}{2}$$



Retrieval costs

Sort by increasing

$$C_0 \times P(R = 0|D = d \wedge Q = q) - C_1 \times P(R = 1|D = d \wedge Q = q)$$

Retrieval costs

Sort by increasing

$$C_0 \times P(R = 0|D = d \wedge Q = q) - C_1 \times P(R = 1|D = d \wedge Q = q)$$

	Return	Not return	
Relevant	0	$-C_1$	$P(R = 1 D = d \wedge Q = q)$
Not relevant	$-C_0$	0	$P(R = 0 D = d \wedge Q = q)$

diff

Retrieval costs

Sort by increasing

$$\underline{C_0} \times P(R = 0|D = d \wedge Q = q) - C_1 \times P(R = 1|D = d \wedge Q = q)$$

Cost of returning an irrelevant document

Retrieval costs

Sort by increasing

$$C_0 \times P(R = 0|D = d \wedge Q = q) - \underline{C_1} \times P(R = 1|D = d \wedge Q = q)$$

Cost of not returning a relevant document

With identical costs...

Sort by increasing

$$C(1 - \times P(R = 1|D = d \wedge Q = q)) - C \times P(R = 1|D = d \wedge Q = q)$$

With identical costs...

Sort by increasing

$$1 - 2 \times P(R = 1 | D = d \wedge Q = q)$$

With identical costs...

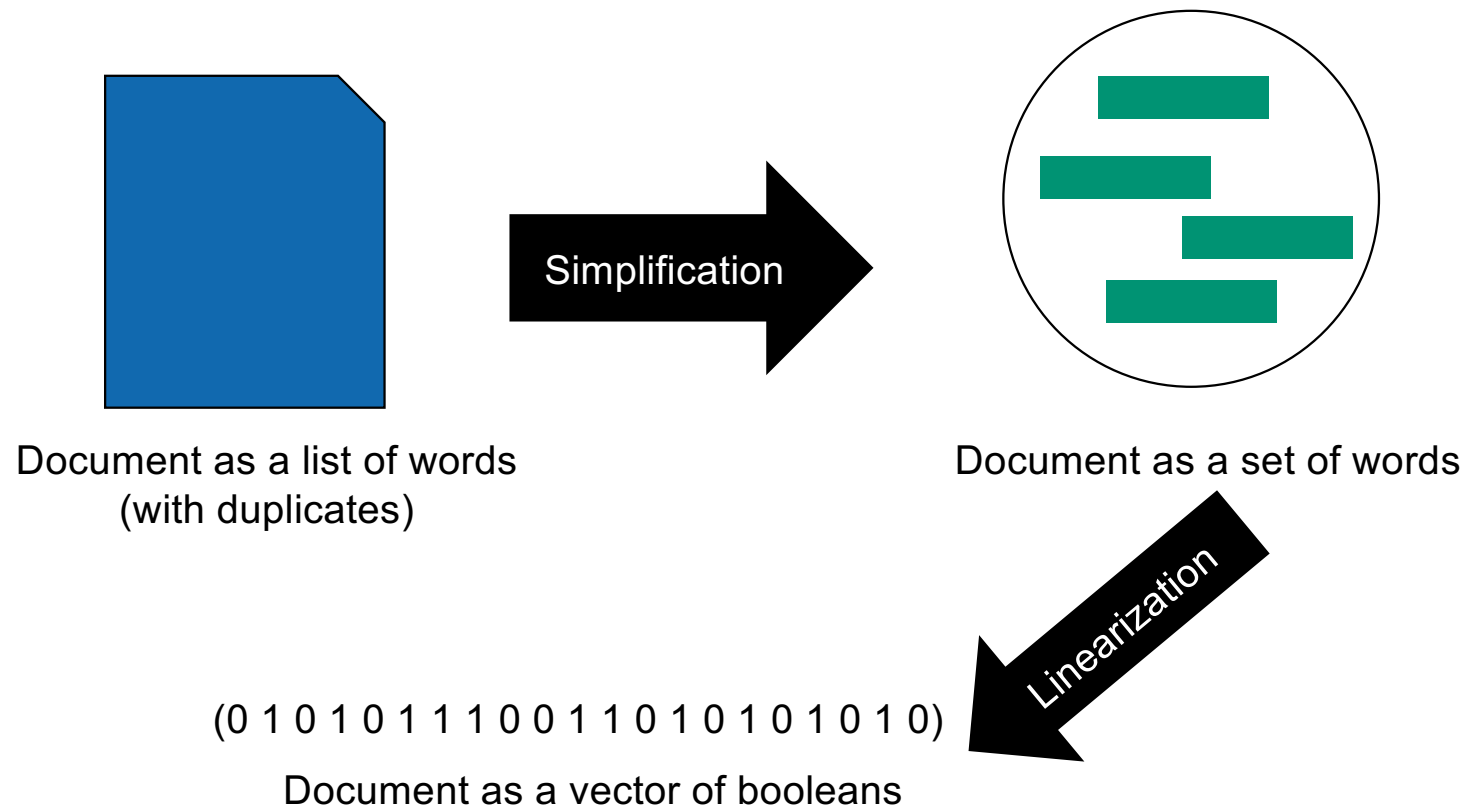
Sort by decreasing

$$P(R = 1 | D = d \wedge Q = q)$$

We "fall back" to the previous method

Binary independence
model

Model and abstraction

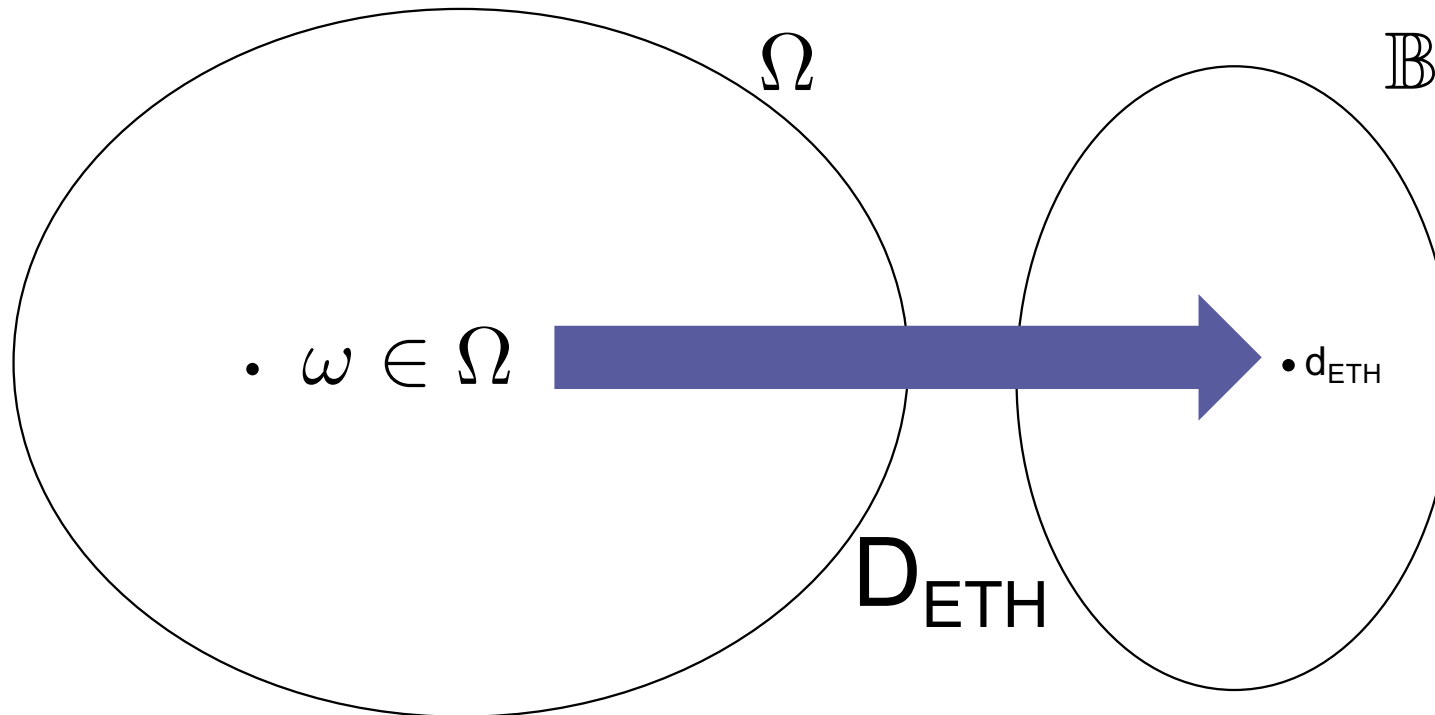


Documents and queries as vectors

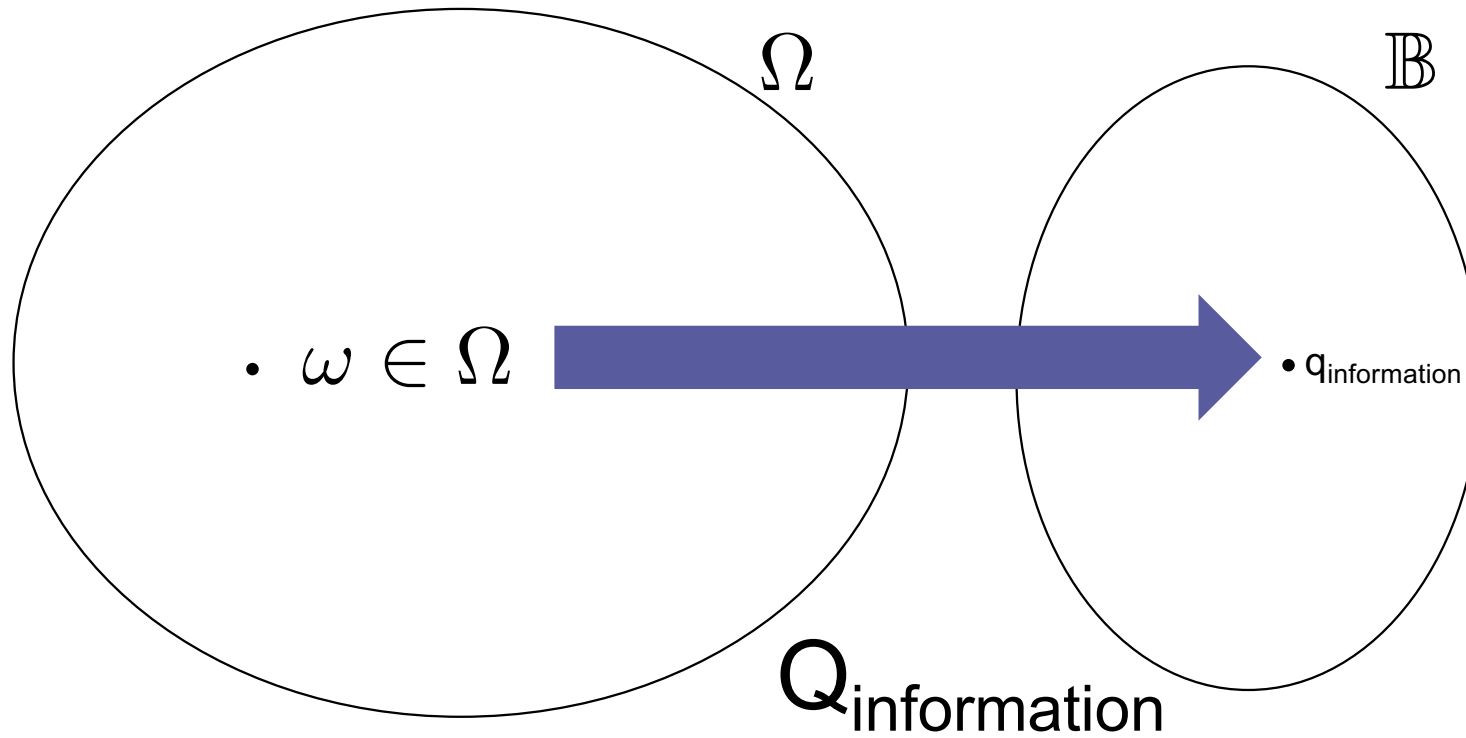
$$d = \begin{pmatrix} d_1 \\ d_2 \\ d_3 \\ \dots \end{pmatrix}$$

$$q = \begin{pmatrix} q_1 \\ q_2 \\ q_3 \\ \dots \end{pmatrix}$$

One "document" random variable per term



One "query" random variable per term



So we can now write things like...

$$P(D_k = d_k | R = 1 \wedge Q = q)$$

So we can now write things like...

$$P(D_k = d_k | R = 1 \wedge Q = q)$$

If $d_k = 1$: Probability that the document d contains term $\mathbf{k}$, knowing that the query is $\mathbf{q}$ and that it is relevant.

If $d_k = 0$: Probability that the document does not contain term $\mathbf{k}$, knowing that the query is $\mathbf{q}$ and that it is relevant.

Naive Bayes Assumption

$$P(D = d) =$$

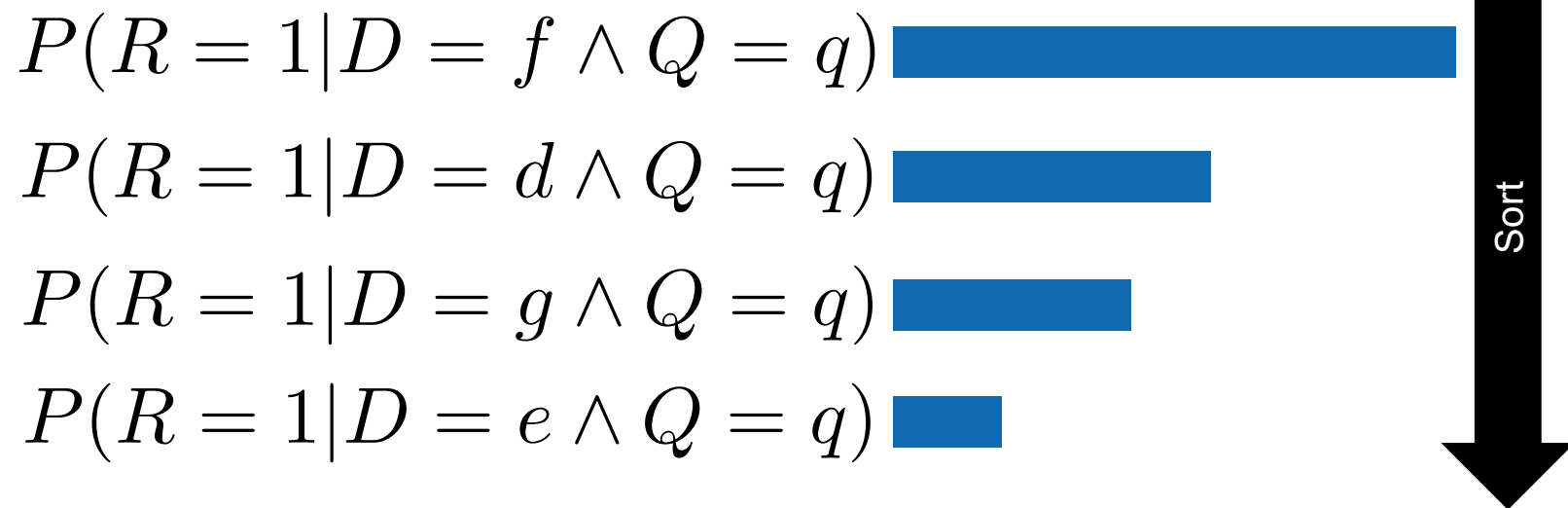
Term occur "independently" in documents

Naive Bayes Assumption

$$P(D = d) = \prod_{k=1}^{k=M} P(D_k = d_k)$$

Term occur "independently" in documents

Going back to what we want to rank...



Bayes' formula

$$P(R = 1|D = d) = \frac{P(D = d|R = 1) \times P(R = 1)}{P(D = d)}$$

Condition on a query q ...

$$P(R = 1|D = d \wedge Q = q) = \frac{P(D = d|R = 1 \wedge Q = q) \times P(R = 1|Q = q)}{P(D = d|Q = q)}$$

Condition on a query q ...

$$P(R = 1|D = d \wedge Q = q) = \frac{P(D = d|R = 1 \wedge Q = q) \times P(R = 1|Q = q)}{P(D = d|Q = q)}$$

That's a lot to evaluate!

Condition on a query q ...

$$P(R = 1|D = d \wedge Q = q) = \frac{P(D = d|R = 1 \wedge Q = q) \times P(R = 1|Q = q)}{P(D = d|Q = q)}$$

Can we get rid of some of it?

Condition on a query q ...

$$P(R = 1|D = d \wedge Q = q) = \frac{P(D = d|R = 1 \wedge Q = q) \times P(R = 1|Q = q)}{P(D = d|Q = q)}$$

Condition on a query q ...

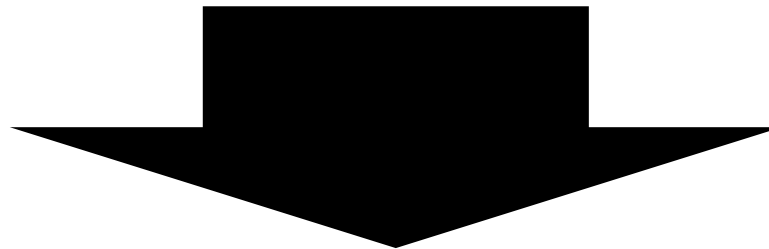
$$P(R = 1|D = d \wedge Q = q) = \frac{P(D = d|R = 1 \wedge Q = q) \times P(R = 1|Q = q)}{P(D = d|Q = q)}$$

$$P(R = 0|D = d \wedge Q = q) = \frac{P(D = d|R = 0 \wedge Q = q) \times P(R = 0|Q = q)}{P(D = d|Q = q)}$$

Condition on a query q ...

$$P(R = 1|D = d \wedge Q = q) = \frac{P(D = d|R = 1 \wedge Q = q) \times P(R = 1|Q = q)}{P(D = d|Q = q)}$$

$$P(R = 0|D = d \wedge Q = q) = \frac{P(D = d|R = 0 \wedge Q = q) \times P(R = 0|Q = q)}{P(D = d|Q = q)}$$

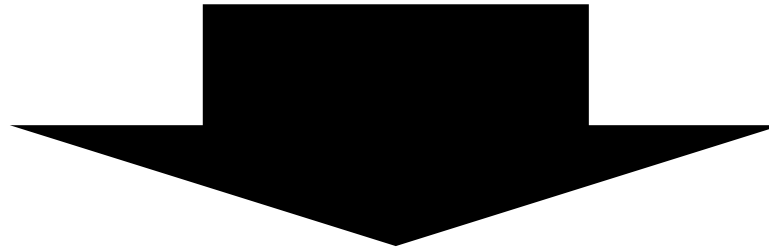


$$O(R = 1|D = d \wedge Q = q) = \frac{P(R = 1|D = d \wedge Q = q)}{P(R = 0|D = d \wedge Q = q)}$$

Condition on a query q ...

$$P(R = 1|D = d \wedge Q = q) = \frac{P(D = d|R = 1 \wedge Q = q) \times P(R = 1|Q = q)}{P(D = d|Q = q)}$$

$$P(R = 0|D = d \wedge Q = q) = \frac{P(D = d|R = 0 \wedge Q = q) \times P(R = 0|Q = q)}{P(D = d|Q = q)}$$

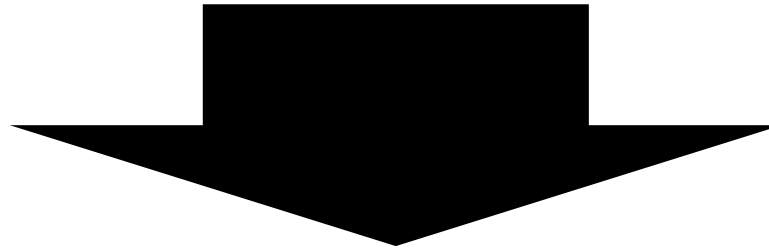


$$O(R = 1|D = d \wedge Q = q) = \frac{P(D = d|R = 1 \wedge Q = q) \times P(R = 1|Q = q)}{P(D = d|R = 0 \wedge Q = q) \times P(R = 0|Q = q)}$$

Condition on a query q ...

$$P(R = 1|D = d \wedge Q = q) = \frac{P(D = d|R = 1 \wedge Q = q) \times P(R = 1|Q = q)}{P(D = d|Q = q)}$$

$$P(R = 0|D = d \wedge Q = q) = \frac{P(D = d|R = 0 \wedge Q = q) \times P(R = 0|Q = q)}{P(D = d|Q = q)}$$



$$O(R = 1|D = d \wedge Q = q) = \frac{P(D = d|R = 1 \wedge Q = q) \times P(R = 1|Q = q)}{P(D = d|R = 0 \wedge Q = q) \times P(R = 0|Q = q)}$$

Condition on a query q ...

$$O(R = 1|D = d \wedge Q = q) = \frac{P(D = d|R = 1 \wedge Q = q) \times P(R = 1|Q = q)}{P(D = d|R = 0 \wedge Q = q) \times P(R = 0|Q = q)}$$

Condition on a query q ...

$$O(R = 1|D = d \wedge Q = q) = \frac{P(D = d|R = 1 \wedge Q = q)}{P(D = d|R = 0 \wedge Q = q)} \times \frac{P(R = 1|Q = q)}{P(R = 0|Q = q)}$$

Condition on a query q ...

$$O(R = 1|D = d \wedge Q = q) = \frac{P(D = d|R = 1 \wedge Q = q)}{P(D = d|R = 0 \wedge Q = q)} \times \frac{P(R = 1|Q = q)}{P(R = 0|Q = q)}$$

These are odds!

Condition on a query q ...

$$O(R = 1|D = d \wedge Q = q) = \frac{P(D = d|R = 1 \wedge Q = q)}{P(D = d|R = 0 \wedge Q = q)} \times O(R = 1|Q = q)$$

These are odds!

Condition on a query q ...

$$O(R = 1|D = d \wedge Q = q) = \frac{P(D = d|R = 1 \wedge Q = q)}{P(D = d|R = 0 \wedge Q = q)} \times O(R = 1|Q = q)$$

Condition on a query q ...

$$O(R = 1|D = d \wedge Q = q) = \frac{P(D = d|R = 1 \wedge Q = q)}{P(D = d|R = 0 \wedge Q = q)} \times O(R = 1|Q = q)$$

We can use our
independence
assumption
here

Naive Bayes Assumption

$$P(D = d) = \prod_{k=1}^{k=M} P(D_k = d_k)$$

Terms occur "independently" in documents

Naive Bayes Assumption

$$P(D = d | R = 1 \wedge Q = q) = \prod_{k=1}^{k=M} P(D_k = d_k | R = 1 \wedge Q = q)$$

Terms occur "independently" in documents
even conditioned on relevance and a given query

Condition on a query q ...

$$O(R = 1|D = d \wedge Q = q) = \frac{P(D = d|R = 1 \wedge Q = q)}{P(D = d|R = 0 \wedge Q = q)} \times O(R = 1|Q = q)$$

We can use our
independence
assumption
here

Condition on a query q ...

$$O(R = 1|D = d \wedge Q = q) = \frac{\prod_{k=1}^{k=M} P(D_k = d_k|R = 1 \wedge Q = q)}{\prod_{k=1}^{k=M} P(D_k = d_k|R = 0 \wedge Q = q)} \times O(R = 1|Q = q)$$

Condition on a query q ...

$$O(R = 1|D = d \wedge Q = q) = \prod_{k=1}^{k=M} \frac{P(D_k = d_k|R = 1 \wedge Q = q)}{P(D_k = d_k|R = 0 \wedge Q = q)} \times O(R = 1|Q = q)$$

Condition on a query q ...

$$O(R = 1 | D = d \wedge Q = q) = \prod_{k=1}^{k=M} \frac{P(D_k = d_k | R = 1 \wedge Q = q)}{P(D_k = d_k | R = 0 \wedge Q = q)} \times O(R = 1 | Q = q)$$

This is boolean!

Condition on a query q ...

$$O(R = 1|D = d \wedge Q = q) = \prod_{k|d_k=0} \frac{P(D_k = 0|R = 1 \wedge Q = q)}{P(D_k = 0|R = 0 \wedge Q = q)} \times \prod_{k|d_k=1} \frac{P(D_k = 1|R = 1 \wedge Q = q)}{P(D_k = 1|R = 0 \wedge Q = q)} \times O(R = 1|Q = q)$$

Condition on a query q ...

We call this p_{kq}

$$O(R = 1|D = d \wedge Q = q) = \prod_{k|d_k=0} \frac{P(D_k = 0|R = 1 \wedge Q = q)}{P(D_k = 0|R = 0 \wedge Q = q)} \times \prod_{k|d_k=1} \frac{P(D_k = 1|R = 1 \wedge Q = q)}{P(D_k = 1|R = 0 \wedge Q = q)} \times O(R = 1|Q = q)$$

Condition on a query q ...

$$O(R = 1 | D = d \wedge Q = q) = \prod_{k | d_k = 0} \frac{P(D_k = 0 | R = 1 \wedge Q = q)}{P(D_k = 0 | R = 0 \wedge Q = q)} \times \prod_{k | d_k = 1} \frac{P(D_k = 1 | R = 1 \wedge Q = q)}{P(D_k = 1 | R = 0 \wedge Q = q)} \times O(R = 1 | Q = q)$$

We call this p_{kq}

We call this u_{kq}

Condition on a query q ...

$$O(R = 1 | D = d \wedge Q = q) = \prod_{k|d_k=0} \frac{P(D_k = 0 | R = 1 \wedge Q = q)}{P(D_k = 0 | R = 0 \wedge Q = q)} \times \prod_{k|d_k=1} \frac{P(D_k = 1 | R = 1 \wedge Q = q)}{P(D_k = 1 | R = 0 \wedge Q = q)} \times O(R = 1 | Q = q)$$

We call this p_{kq}

We call this u_{kq}

	k^{th} term present	k^{th} term absent
Relevant	p_{kq}	$1-p_{kq}$
Not relevant	u_{kq}	$1-u_{kq}$

Simplifying assumption for terms not in the query

$$q_k = 0 \implies p_{kq} = u_{kq}$$

Condition on a query q ...

$$O(R = 1 | D = d \wedge Q = q) = \prod_{k|d_k=0} \frac{P(D_k = 0 | R = 1 \wedge Q = q)}{P(D_k = 0 | R = 0 \wedge Q = q)} \times \prod_{k|d_k=1} \frac{P(D_k = 1 | R = 1 \wedge Q = q)}{P(D_k = 1 | R = 0 \wedge Q = q)} \times O(R = 1 | Q = q)$$

We call this p_{kq}

We call this u_{kq}

	k^{th} term present	k^{th} term absent
Relevant	p_{kq}	$1-p_{kq}$
Not relevant	u_{kq}	$1-u_{kq}$

Condition on a query q ...

$$O(R = 1|D = d \wedge Q = q) = \prod_{\substack{k|d_k=0 \\ \wedge q_k=1}} \frac{P(D_k = 0|R = 1 \wedge Q = q)}{P(D_k = 0|R = 0 \wedge Q = q)} \times \prod_{\substack{k|d_k=1 \\ \wedge q_k=1}} \frac{P(D_k = 1|R = 1 \wedge Q = q)}{P(D_k = 1|R = 0 \wedge Q = q)} \times O(R = 1|Q = q)$$

We call this p_{kq}

We call this u_{kq}

	k^{th} term present	k^{th} term absent
Relevant	p_{kq}	$1-p_{kq}$
Not relevant	u_{kq}	$1-u_{kq}$

Simplifying assumption for terms in the query

	k^{th} term present	k^{th} term absent
Relevant	p_{kq}	$1-p_{kq}$
Not relevant	u_{kq}	$1-u_{kq}$



	k^{th} term present	k^{th} term absent
Relevant	p_k	$1-p_k$
Not relevant	u_k	$1-u_k$

Condition on a query q ...

$$O(R = 1|D = d \wedge Q = q) = \prod_{\substack{k|d_k=0 \\ \wedge q_k=1}} \frac{P(D_k = 0|R = 1 \wedge Q = q)}{P(D_k = 0|R = 0 \wedge Q = q)} \times \prod_{\substack{k|d_k=1 \\ \wedge q_k=1}} \frac{P(D_k = 1|R = 1 \wedge Q = q)}{P(D_k = 1|R = 0 \wedge Q = q)} \times O(R = 1|Q = q)$$

We call this p_{kq}

We call this u_{kq}

	k^{th} term present	k^{th} term absent
Relevant	p_{kq}	$1-p_{kq}$
Not relevant	u_{kq}	$1-u_{kq}$

Condition on a query q ...

$$O(R = 1 | D = d \wedge Q = q) = \prod_{\substack{k | d_k = 0 \\ \wedge q_k = 1}} \frac{P(D_k = 0 | R = 1 \wedge Q = q)}{P(D_k = 0 | R = 0 \wedge Q = q)} \times \prod_{\substack{k | d_k = 1 \\ \wedge q_k = 1}} \frac{P(D_k = 1 | R = 1 \wedge Q = q)}{P(D_k = 1 | R = 0 \wedge Q = q)} \times O(R = 1 | Q = q)$$

We call this p_k

We call this u_k

	k^{th} term present	k^{th} term absent
Relevant	p_k	$1 - p_k$
Not relevant	u_k	$1 - u_k$

Condition on a query q ...

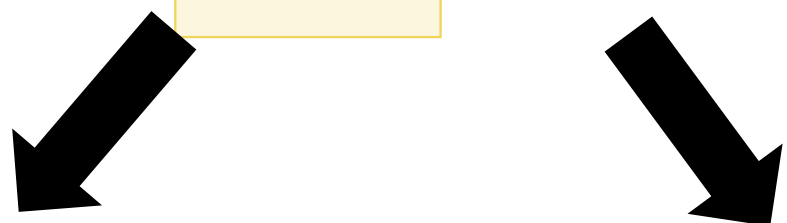
$$O(R = 1|D = d \wedge Q = q) = \prod_{k|d_k=1 \wedge q_k=1} \frac{p_k}{u_k} \times \prod_{k|d_k=0 \wedge q_k=1} \frac{1-p_k}{1-u_k} \times O(R = 1|Q = q)$$

Condition on a query q ...

$$O(R = 1|D = d \wedge Q = q) = \prod_{k|d_k=1 \wedge q_k=1} \frac{p_k}{u_k} \times \prod_{k|d_k=0 \wedge q_k=1} \frac{1-p_k}{1-u_k} \times O(R = 1|Q = q)$$

Terms found in d

Condition on a query q ...

$$O(R = 1|D = d \wedge Q = q) = \prod_{k|d_k=1 \wedge q_k=1} \frac{p_k}{u_k} \times \prod_{k|d_k=0 \wedge q_k=1} \frac{1-p_k}{1-u_k} \times O(R = 1|Q = q)$$


divide with $\prod_{k|d_k=1 \wedge q_k=1} \frac{1-p_k}{1-u_k}$

multiply with $\prod_{k|d_k=1 \wedge q_k=1} \frac{1-p_k}{1-u_k}$

Condition on a query q ...

$$O(R = 1|D = d \wedge Q = q) = \prod_{k|d_k=1 \wedge q_k=1} \frac{p_k \times (1 - u_k)}{u_k \times (1 - p_k)} \times \prod_{k|q_k=1} \frac{1 - p_k}{1 - u_k} \times O(R = 1|Q = q)$$

Condition on a query q ...

$$O(R = 1|D = d \wedge Q = q) = \prod_{k|d_k=1 \wedge q_k=1} \frac{p_k \times (1 - u_k)}{u_k \times (1 - p_k)} \times \prod_{k|q_k=1} \frac{1 - p_k}{1 - u_k} \times O(R = 1|Q = q)$$

This does not depend on d !

Condition on a query q ...

$$O(R = 1|D = d \wedge Q = q) = \prod_{k|d_k=1 \wedge q_k=1} \frac{p_k \times (1 - u_k)}{u_k \times (1 - p_k)} \times \prod_{k|q_k=1} \frac{1 - p_k}{1 - u_k} \times O(R = 1|Q = q)$$

This does not depend on d !



Condition on a query q ...

$$O(R = 1|D = d \wedge Q = q) = \prod_{k|d_k=1 \wedge q_k=1} \frac{p_k \times (1 - u_k)}{u_k \times (1 - p_k)} \times \prod_{k|q_k=1} \frac{1 - p_k}{1 - u_k} \times O(R = 1|Q = q)$$

This is what we care about!

This does not depend on d !



Condition on a query q ...

$$O(R = 1 | D = d \wedge Q = q) = \prod_{k | d_k = 1 \wedge q_k = 1} \frac{p_k \times (1 - u_k)}{u_k \times (1 - p_k)}$$

Condition on a query q ...

$$O(R = 1 | D = d \wedge Q = q) = \prod_{k | d_k = 1 \wedge q_k = 1} \frac{p_k \times (1 - u_k)}{u_k \times (1 - p_k)} \longrightarrow \log$$

Odds ratio

$$RSV_d = \log \prod_{k|d_k=1 \wedge q_k=1} \frac{\frac{p_k}{1-p_k}}{\frac{u_k}{1-u_k}}$$

Retrieval Status Value

$$RSV_d = \sum_{k|d_k=1 \wedge q_k=1} \log \frac{p_k}{1-p_k} - \log \frac{u_k}{1-u_k}$$

Retrieval Status Value

$$RSV_d = \sum_{k|d_k=1 \wedge q_k=1} \log \frac{p_k}{1 - p_k} - \log \frac{u_k}{1 - u_k}$$

Odds of containing term k
in relevant documents

Retrieval Status Value

$$RSV_d = \sum_{k|d_k=1 \wedge q_k=1} \log \frac{p_k}{1 - p_k} - \log \frac{u_k}{1 - u_k}$$

Odds of containing term k
in relevant documents

Odds of containing term k
in non-relevant documents

Retrieval Status Value

$$RSV_d = \sum_{k|d_k=1 \wedge q_k=1} \log \frac{p_k}{1-p_k} - \log \frac{u_k}{1-u_k}$$

	k th term present	k th term absent
Relevant	p_k	$1-p_k$
Not relevant	u_k	$1-u_k$

RSV as a sum of weights

$$RSV_d = \sum_{k|d_k=1 \wedge q_k=1} c_k$$

RSV as a scalar product

$$RSV_d = \vec{c} \cdot \vec{s}$$

Retrieval Status Value

Weights!

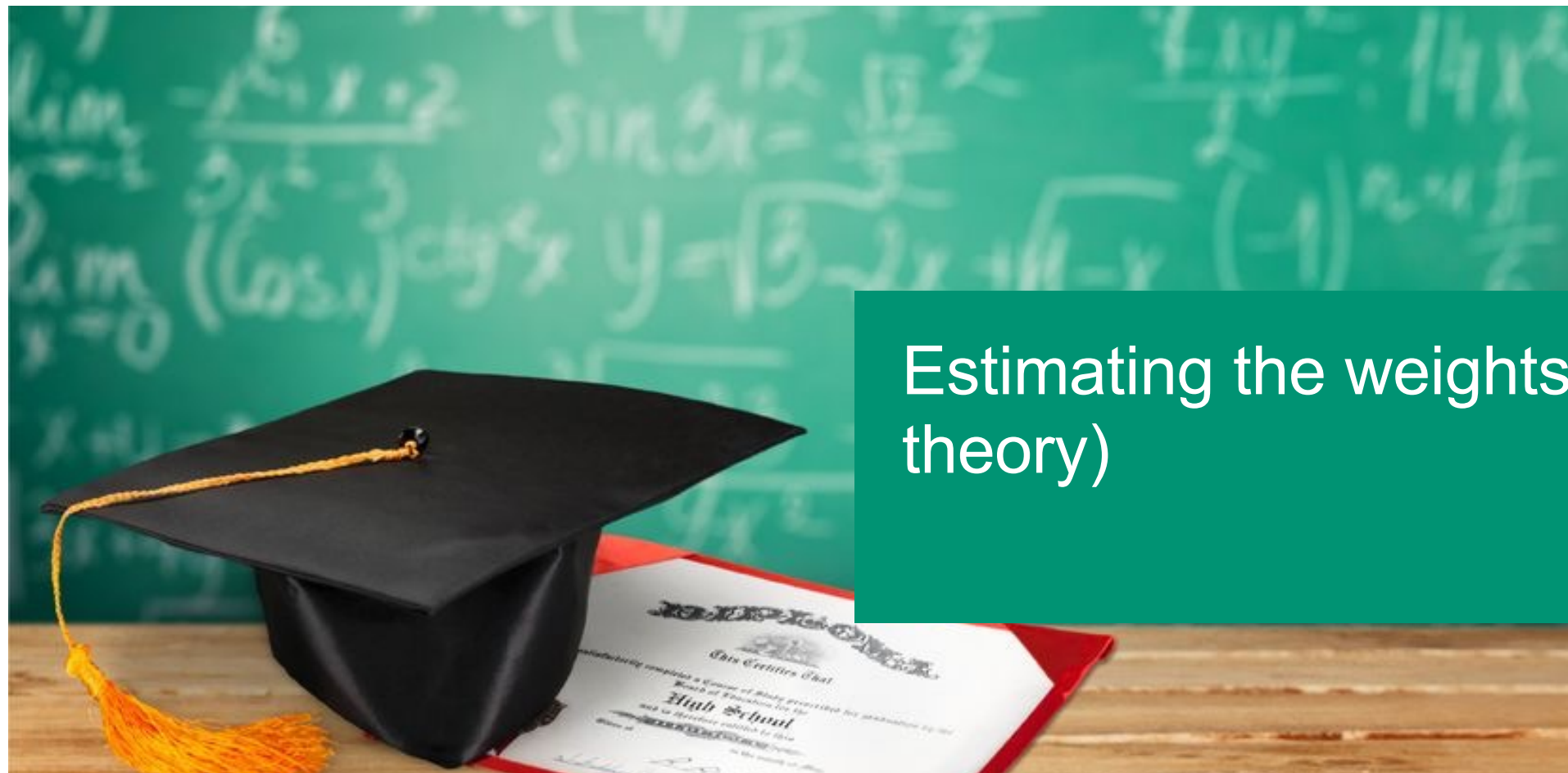
$$RSV_d = \boxed{\vec{c}} \cdot \vec{s}$$

Reminder: Zone queries



Score of a document:

$$\vec{g} \cdot \vec{s}$$



Estimating the weights (in theory)

Contingency table

	k^{th} term present	k^{th} term absent
Relevant	p_k	$1-p_k$
Not relevant	u_k	$1-u_k$

Contingency table

	k^{th} term present	k^{th} term absent	Total
Relevant			
Not relevant			
Total			N

Contingency table

	k th term present	k th term absent	Total
Relevant			
Not relevant			
Total	df _k		N

Contingency table

	k^{th} term present	k^{th} term absent	Total
Relevant			S
Not relevant			
Total	df_k		N

Contingency table

	k^{th} term present	k^{th} term absent	Total
Relevant			S
Not relevant			N-S
Total	df_k	$N - df_k$	N

Contingency table

	k^{th} term present	k^{th} term absent	Total
Relevant	s	$S-s$	S
Not relevant	$df_k - s$		$N-S$
Total	df_k	$N - df_k$	N

Contingency table

	k^{th} term present	k^{th} term absent	Total
Relevant	s	$S-s$	S
Not relevant	$df_k - s$	$N - df_k - S + s$	$N-S$
Total	df_k	$N - df_k$	N

Contingency table

	k^{th} term present	k^{th} term absent
Relevant	s	$S-s$
Not relevant	$df_k - s$	$N - df_k - S + s$

Contingency table

	k^{th} term present	k^{th} term absent
Relevant	s	$S-s$
Not relevant	$df_k - s$	$N - df_k - S + s$

Odds for p_t

Odds for u_t

Contingency table

	k^{th} term present	k^{th} term absent
Relevant	s	$S-s$
Not relevant	$df_k - s$	$N - df_k - S + s$

Odds for p_t

Odds for u_t

$$c = \log \frac{p_k}{1 - p_k} + \log \frac{1 - u_k}{u_k}$$

Contingency table

	k th term present	k th term absent
Relevant	s	S-s
Not relevant	df _k - s	N - df _k - S + s

$$C_k = \frac{\frac{s}{S-s}}{\frac{df_k - s}{N - df_k - S + s}}$$

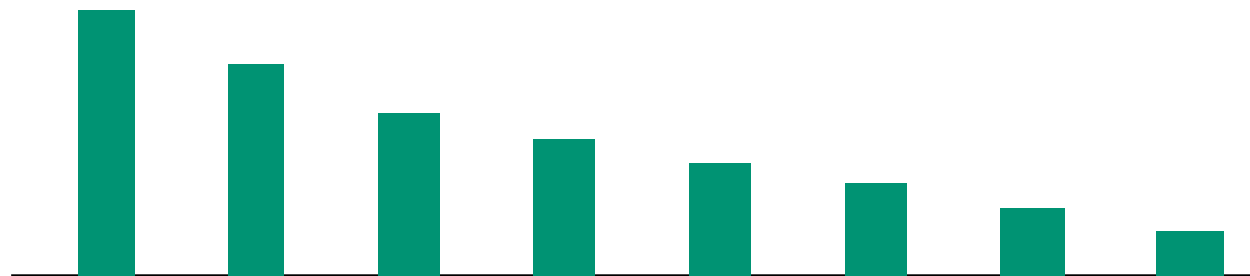
With smoothing

	k th term present	k th term absent
Relevant	s +1/2	S-s +1/2
Not relevant	df _k - s +1/2	N - df _k - S + s +1/2

$$C_k = \frac{\frac{s + \frac{1}{2}}{S - s + \frac{1}{2}}}{\frac{df_k - s + \frac{1}{2}}{N - df_k - S + s + \frac{1}{2}}}$$

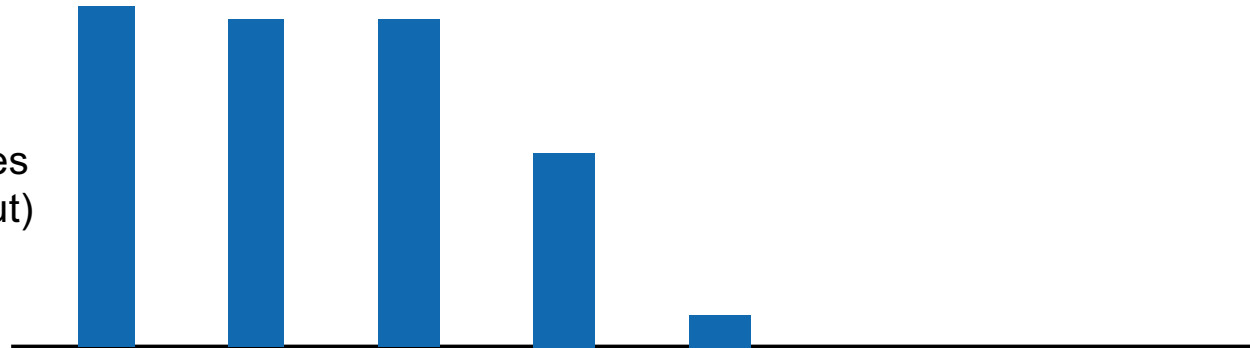
Why smoothing?

Exact values
(impossible
in practice)

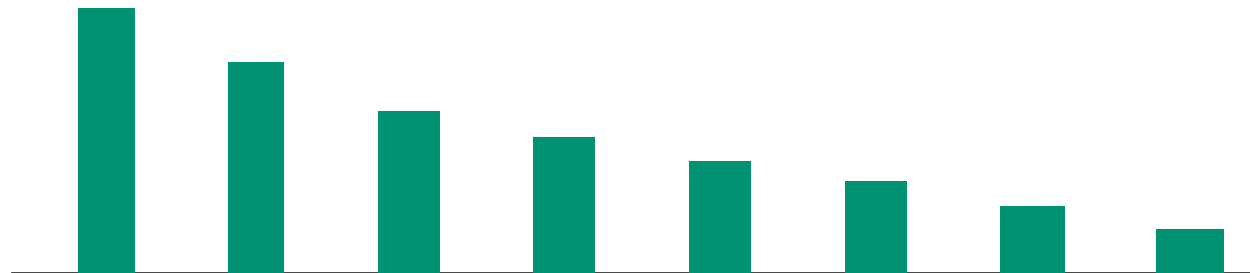


Why smoothing?

Measured values
(on smaller input)

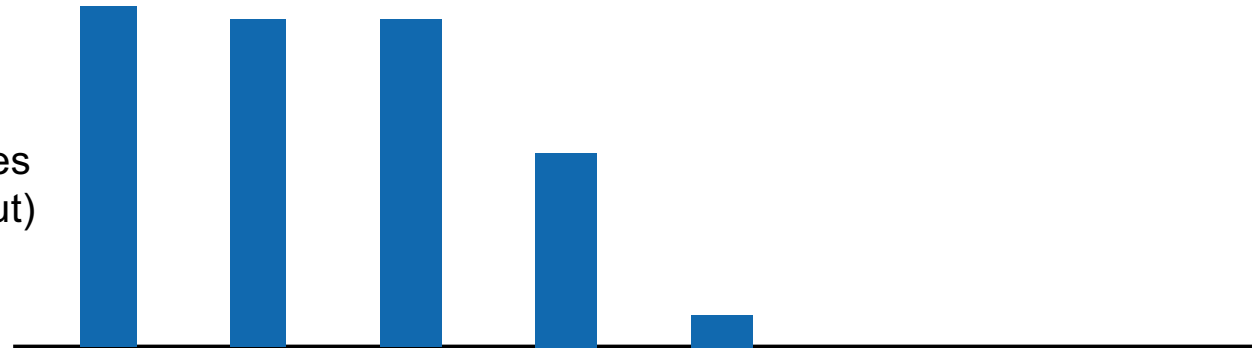


Exact values
(impossible
in practice)



Why smoothing?

Measured values
(on smaller input)



Smoothed values





Estimating the weights (in practice)

Retrieval Status Value

$$RSV_d = \sum_{k|d_k=1 \wedge q_k=1} \log \frac{p_k}{1-p_k} - \log \frac{u_k}{1-u_k}$$

Retrieval Status Value

$$RSV_d = \sum_{k|d_k=1 \wedge q_k=1} \log \frac{p_k}{1 - p_k} - \log \frac{u_k}{1 - u_k}$$

Odds of containing term k
in relevant documents

Retrieval Status Value

$$RSV_d = \sum_{k|d_k=1 \wedge q_k=1} \log \frac{p_k}{1 - p_k} - \log \frac{u_k}{1 - u_k}$$

Croft and Harper suggest
that the odds are 1.

Retrieval Status Value

$$RSV_d = \sum_{k|d_k=1 \wedge q_k=1} \log \frac{p_k}{1 - p_k} - \log \frac{u_k}{1 - u_k}$$

Odds of containing term k
in non-relevant documents

Retrieval Status Value

$$RSV_d = \sum_{k|d_k=1 \wedge q_k=1} \log \frac{p_k}{1 - p_k} - \log \frac{u_k}{1 - u_k}$$

$$\frac{df_k}{N - df_k} \text{ Statistics-based estimate}$$

Odds of containing term k
in non-relevant documents

Retrieval Status Value

$$RSV_d = \sum_{k|d_k=1 \wedge q_k=1} \log 1 - \log \frac{df_k}{N - df_k}$$

Retrieval Status Value

$$RSV_d = \sum_{k|d_k=1 \wedge q_k=1} \log 1 + \log \frac{N - df_k}{df_k}$$

Retrieval Status Value

$$RSV_d = \sum_{k|d_k=1 \wedge q_k=1} \log \frac{N}{df_k}$$

(approximation)

Retrieval Status Value

$$RSV_d = \sum_{k|d_k=1 \wedge q_k=1} \log \frac{N}{df_k}$$

This is the inverted document frequency!

Retrieval Status Value

$$RSV_d = \sum_{k|d_k=1 \wedge q_k=1} \log \frac{N}{df_k}$$

This justifies idf weighting in the Vector-Space Model!



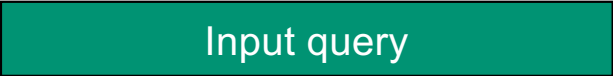
Relevance feedback
(Okapi BM25)

Retrieval Status Value

$$RSV_d = \sum_{k|d_k=1 \wedge q_k=1} \log \frac{p_k}{1 - p_k} - \log \frac{u_k}{1 - u_k}$$

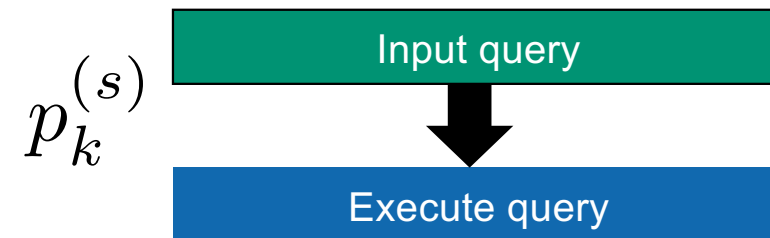
Making it 1
is a bit exaggerated

Relevance feedback

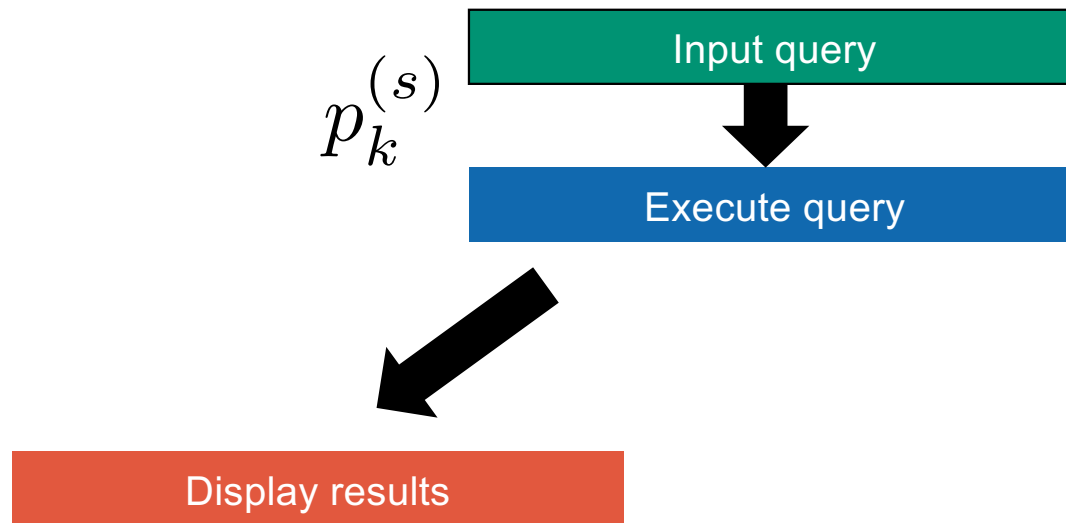


Input query

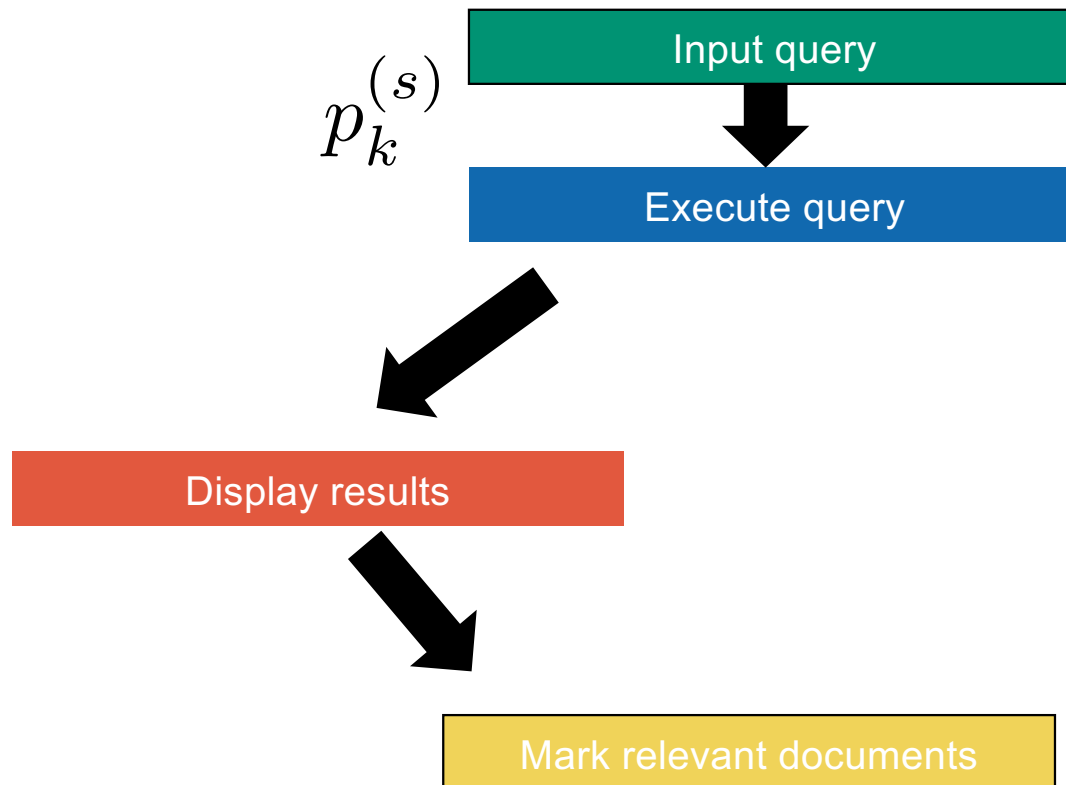
Relevance feedback



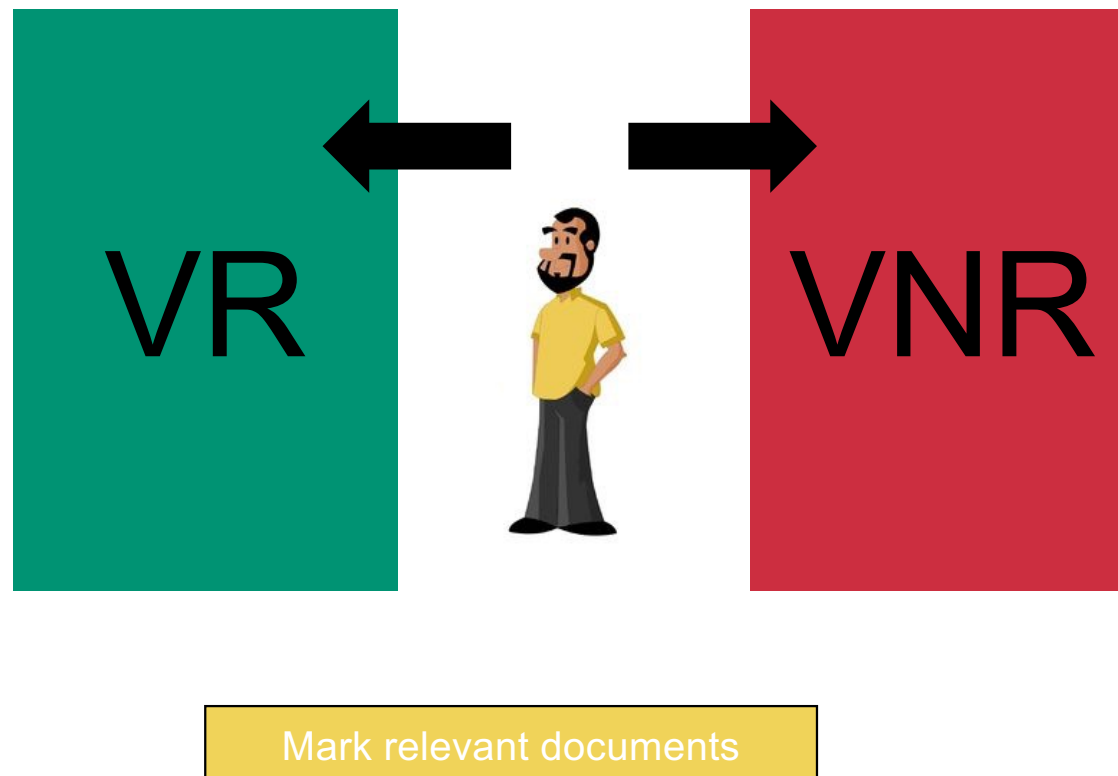
Relevance feedback



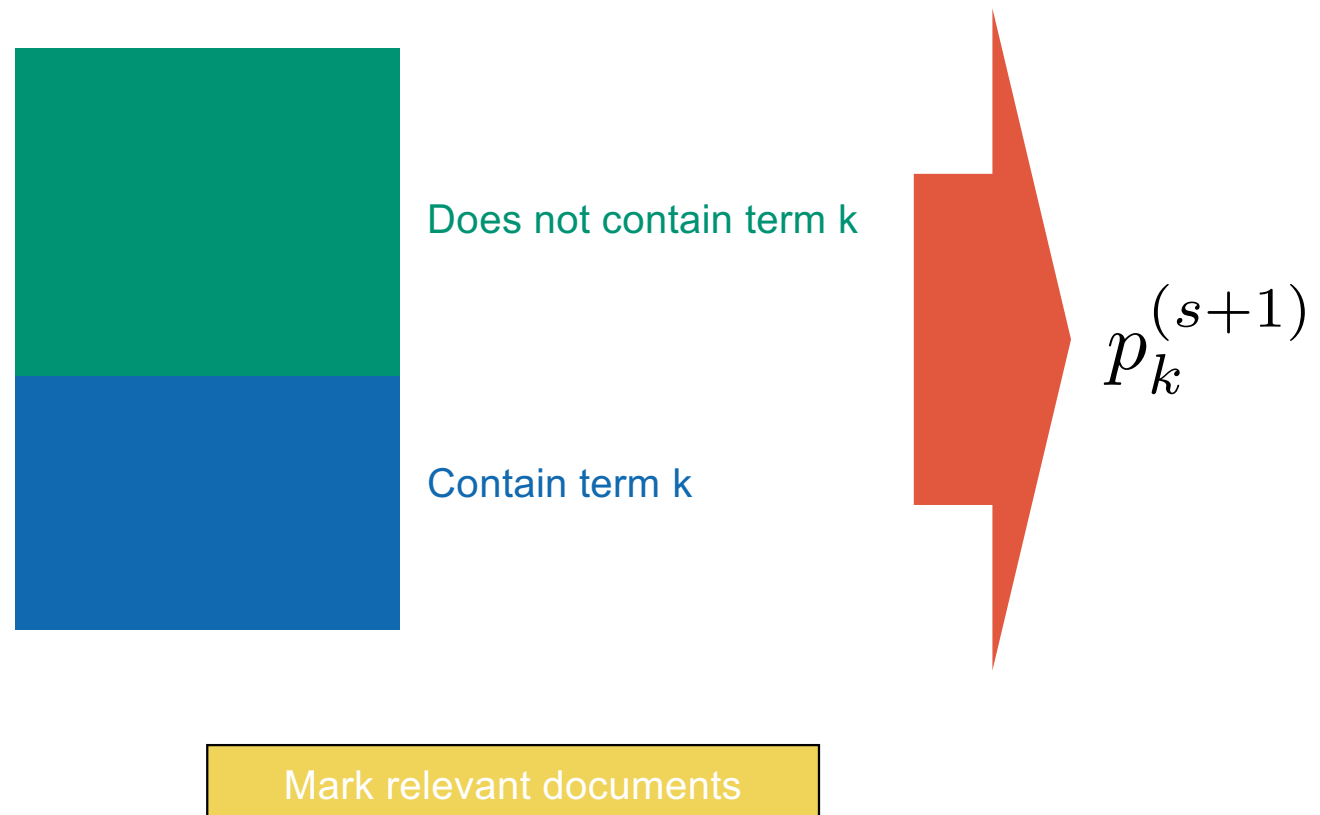
Relevance feedback



Relevance feedback



Relevance feedback



Updating

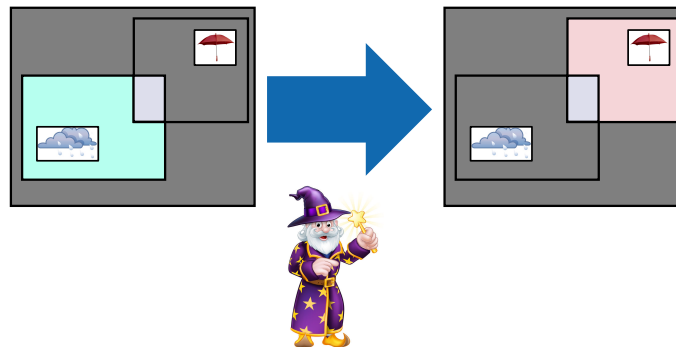
$$p_k^{(s+1)} = \frac{|VR_k|}{|VR|}$$

Smoothed updating

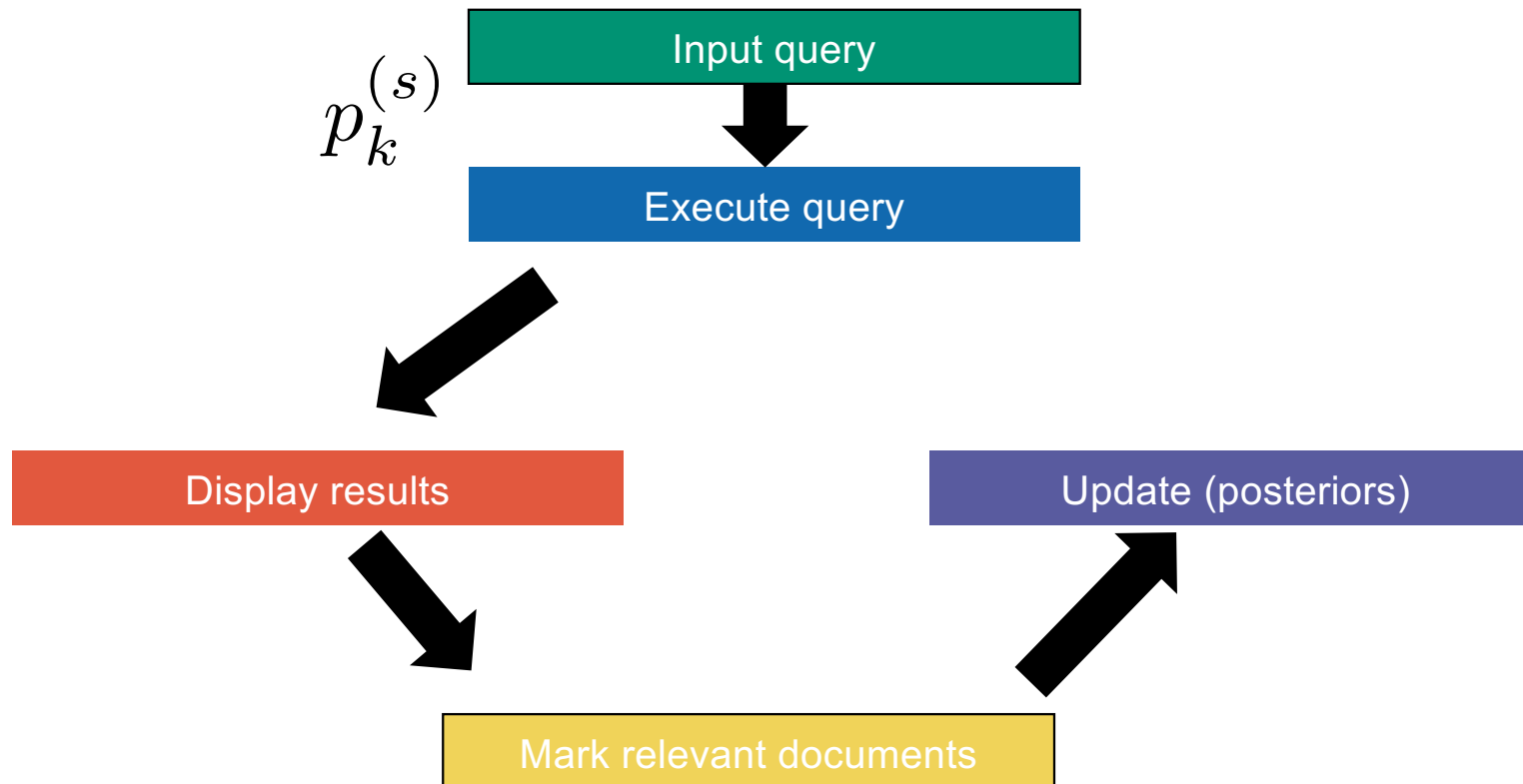
$$p_k^{(s+1)} = \frac{|VR_k| + \frac{1}{2}}{|VR| + 1}$$

Bayesian updating

$$p_k^{(s+1)} = \frac{|VR_k| + \kappa p_k^{(s)}}{|VR| + \kappa}$$



Relevance feedback



Relevance feedback

